



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Multimodal Large Language Models

INFS 4205/7205 Advanced Techniques for High Dimensional Data

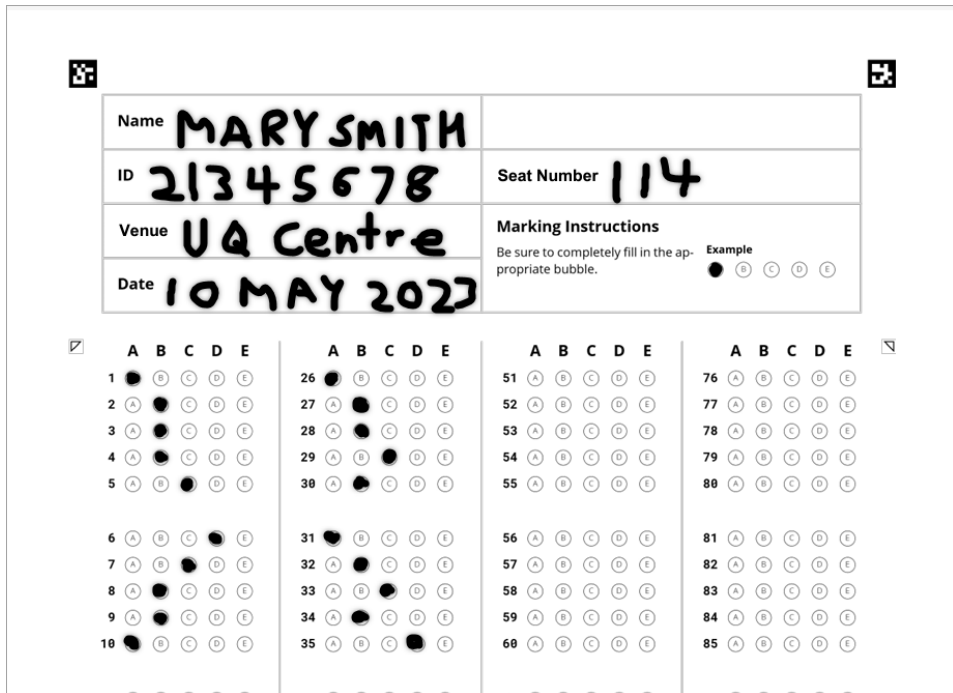
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Assessment Updates

- **A2 In-Semester Quiz**
 - Lecture (4:00pm – 4:30pm) + break 10min
 - Quiz (4:50pm – 5:50pm)
 - MCQ Questions
 - Week 2-5; Closed-book
 - Bubble sheet (bring **2B pencil**)
 - Please bring your **student ID card** 😊

Gradescope Bubble Sheet tips



The bubble sheet template includes a header section for student information and a main section for answers. The header section contains fields for Name, ID, Seat Number, Venue, and Date. The main section consists of a grid of bubbles for answers, with columns labeled A, B, C, D, and E. The bubbles are numbered 1 through 85. The example shows the following information filled in:

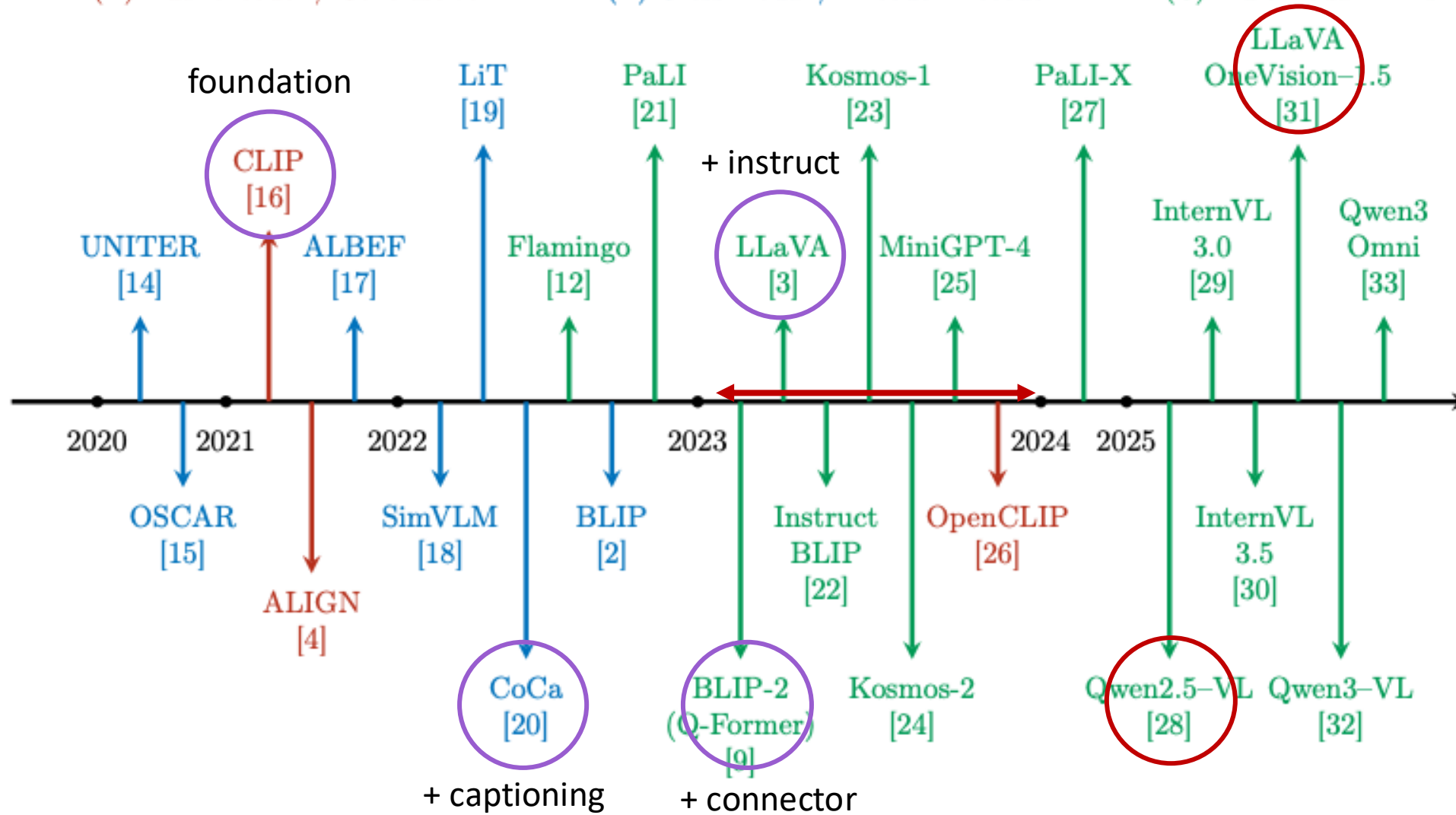
Name	MARY SMITH
ID	21345678
Seat Number	114
Venue	UQ Centre
Date	10 MAY 2023

Marking Instructions: Be sure to completely fill in the appropriate bubble. Example: ● ○ ○ ○ ○

1. Your exam will use a Gradescope Bubble Sheet
2. Enter your **FIRST NAME & LAST NAME** in block capitals
3. Enter your eight digit UQ student **ID**
4. For central exams enter the venue and your seat number
5. Enter the date of your exam
6. Fill in the bubbles using a **2B pencil**
7. To change an answer use an **eraser**

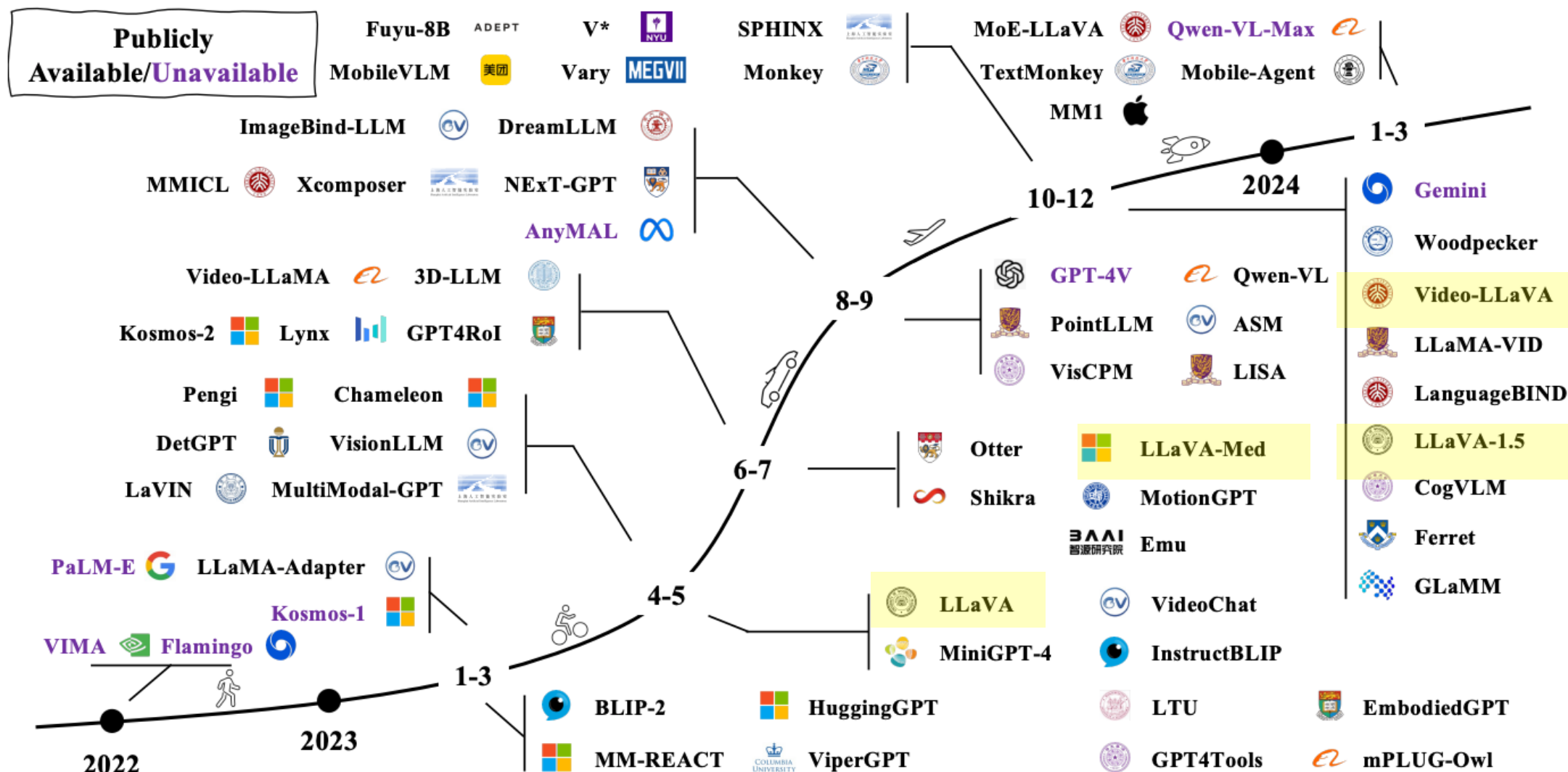
Recap: The Roadmap of VLMs

— (A) Dual-encoder / Contrastive — (B) Cross-modal / Encoder-Decoder — (C) Multimodal LLMs

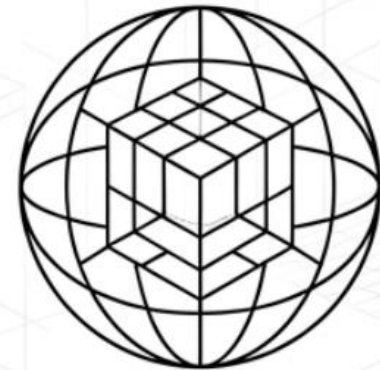
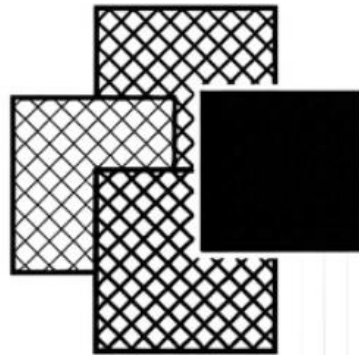
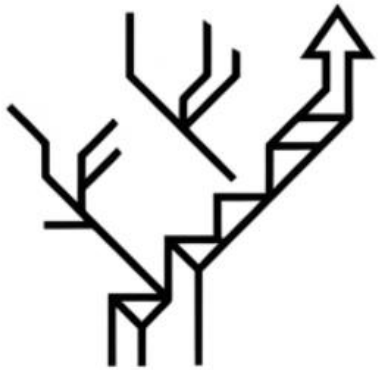


If we take a **closer** look at the advancements made during 2023–2024, ...

Recap: The Roadmap of VLMs



Today's Roadmap



1.

The Evolution of LLaVA: Tracing the lineage from 1.0 to NeXT and Qwen-VL



2.

Unified Multimodal Frameworks:
Architectures that span single-image, multi-image, and video



3.

Multimodal Reasoning & CoT:
Bringing structured logic to visual spaces



4.

A Glimpse into World Models:
Beyond perception into physical world understanding

Recap: LLaVA 1.0

Traditional Captioning: Static, non-interactive, and syntactically shallow. (e.g., CoCa, BLIP-2)



A boy rides a red bicycle in a field.

Conversations: 58K
Multimodal dialogues

Detailed Descriptions: 23K
Rich visual narratives

Complex Reasoning: 77K
Logical inference & understanding

Instruction Tuning: GPT-4 was fed symbolic image data (bounding boxes) to generate 158K rich, multi-turn reasoning conversations. Vision models learned to reason, not just label.

```
messages = [{"role": "user", "content": "I am looking at a single image. What are you looking at. Answer in one word."}]
```

Design a conversation between you and a visual AI assistant to generate a tone that a visual AI assistant can use to respond to your question and give corresponding

Include questions asking
the objects, object act
questions that have def
(1) one can see the cont
(2) one can determine c
that cannot be answer



ssistant, and you are
the same image you

Answers should be in a
ask diverse questions

ject types, counting
ts, etc. Only include

answer confidently;
not ask any question

User: What safety

User: What safety precautions is the rider ignoring?

mage, for example, asking about events happening in

User: Describe the

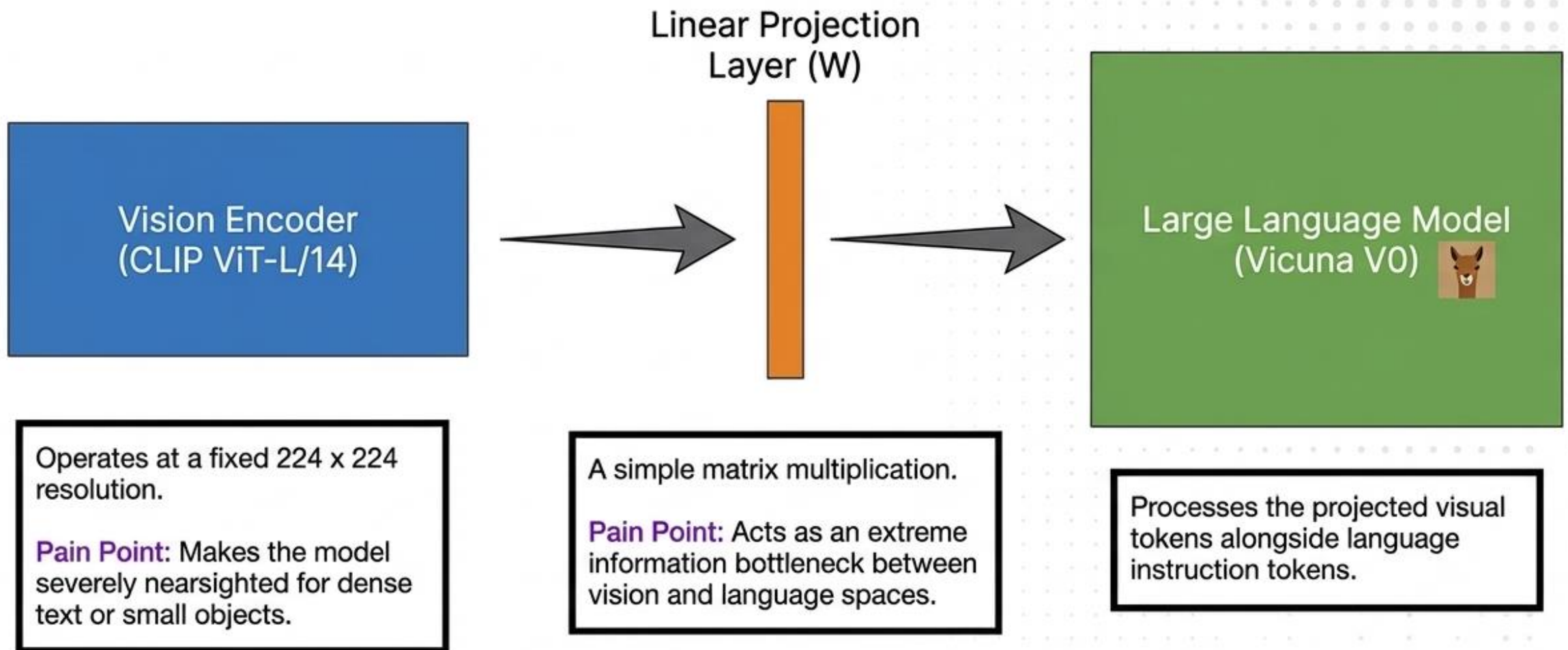
User: Describe the bicycle's condition.

...and answers when answering
...s to make the content more
...cessary.¹⁰⁰¹

```
for sample in fews:
    messages.append(sample)
    messages.append(sample)
messages.append({"
```

AI: The red bicycle appears to be in functional condition, though the frame shows some wear. It is a single-speed model with coaster brakes, suitable for the dirt path.

Recap: LLaVA 1.0



LLaVA 1.5 Problem Statement

Original LLaVA

Method	LLM	Res.	GQA	MME	MM-Vet
InstructBLIP	14B	224	49.5	1212.8	25.6
<i>Only using a subset of InstructBLIP training data</i>					
0 LLaVA	7B	224	–	809.6	25.5
1 +VQA-v2	7B	224	47.0	1197.0	27.7
2 +Format prompt	7B	224	46.8	1323.8	26.3
3 +MLP VL connector	7B	224	47.3	1355.2	27.8
4 +OKVQA/OCR	7B	224	50.0	1377.6	29.6

- Excelled at conversation-style visual reasoning
- Fell short in traditional visual-QA scenarios
- Like other VLMs developed around this time, it only performed well on a specific type of task (e.g., limited to CoCo-GPT4)

LLaVA 1.5: (1) format prompt; (2) connector; (3) more data

Goal: Build a stronger general-purpose visual assistant that is designed with a larger variety of baselines and robust benchmarks

LLaVA 1.5: Power of Prompt

Visual input example, Multitask Balancing Problem:



User Is this unusual? Please explain in detail.
InstructBLIP yes

(a) Example of InstructBLIP [13] (Vicuna-13B) having difficulty balancing between short- and long-form answers.

Visual input example, Different Format Prompts:

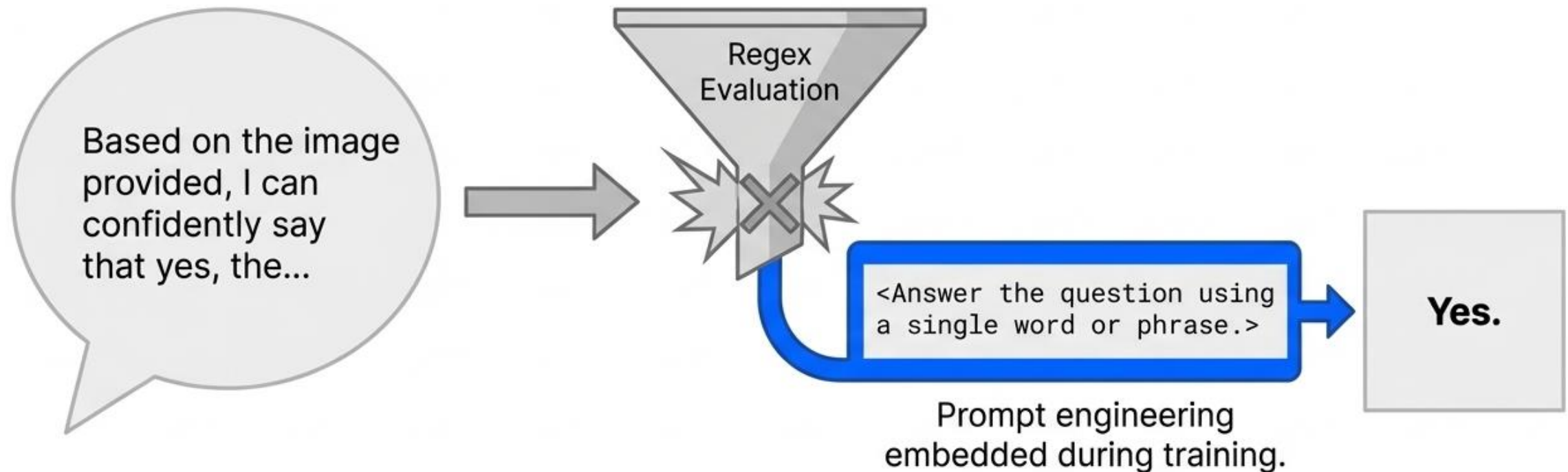
Normal prompt	What is the color of the shirt that the man is wearing?
Response	The man is wearing a yellow shirt.
Ambiguous prompt	Q: What is the color of the shirt that the man is wearing? A:
Response	The man is wearing a yellow shirt.
Formatting prompt	What is the color of the shirt that the man is wearing? Answer the question using a single word or phrase.
Response	Yellow.

(b) Comparison of how different prompts regularize the output format. The results are obtained zero-shot directly after LLaVA undergoes the first-stage vision-language alignment pretraining, without the second-stage visual instruction tuning.

Specific prompt -> specific answer

LLaVA 1.5: Power of Prompt

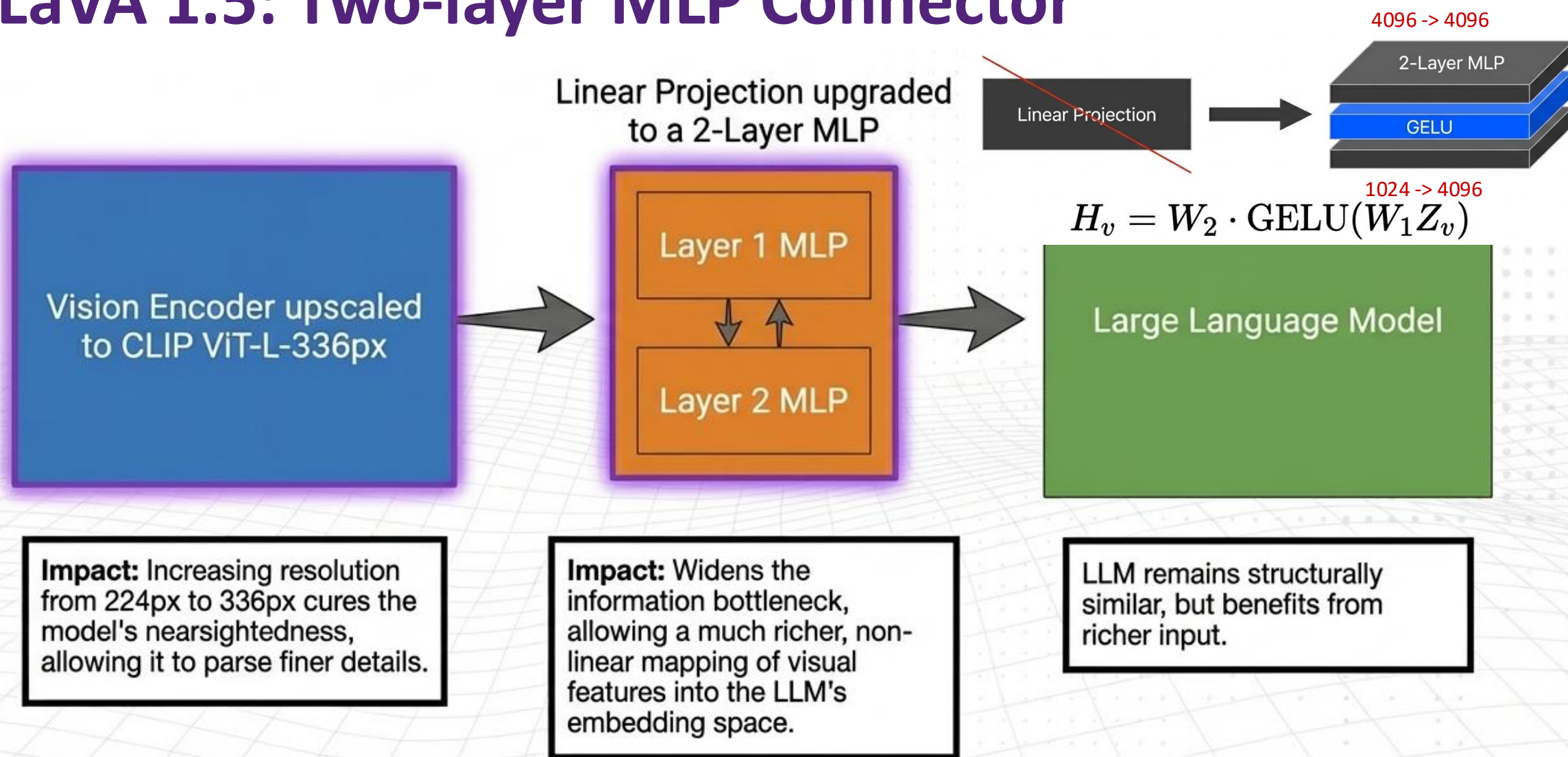
Data Mixing Strategy: Fine-tuned LLMs naturally become conversational. In academic VQA benchmarks (where the ground truth is simply "Yes"), verbosity causes regex evaluation scripts to fail.



When the model's outputs are lengthy and free-form...

Prevent over-rambling

LLaVA 1.5: Two-layer MLP Connector



LLaVA 1.5: More Data

VQA-v2 and GQA

- General academic tasks (open-ended short answer, true/false etc.)
- Prompt + “answer the question using a single word or phrase”

OKVQA

- Prompt + “answer the option’s letter from the given choices directly”

OCRVQA

- Images contain text vital to academic task

ShareGPT

- Allows for multilingual abilities (Spanish, Japanese, Korean, Chinese etc)

What task is the man performing?
talking on phone eating

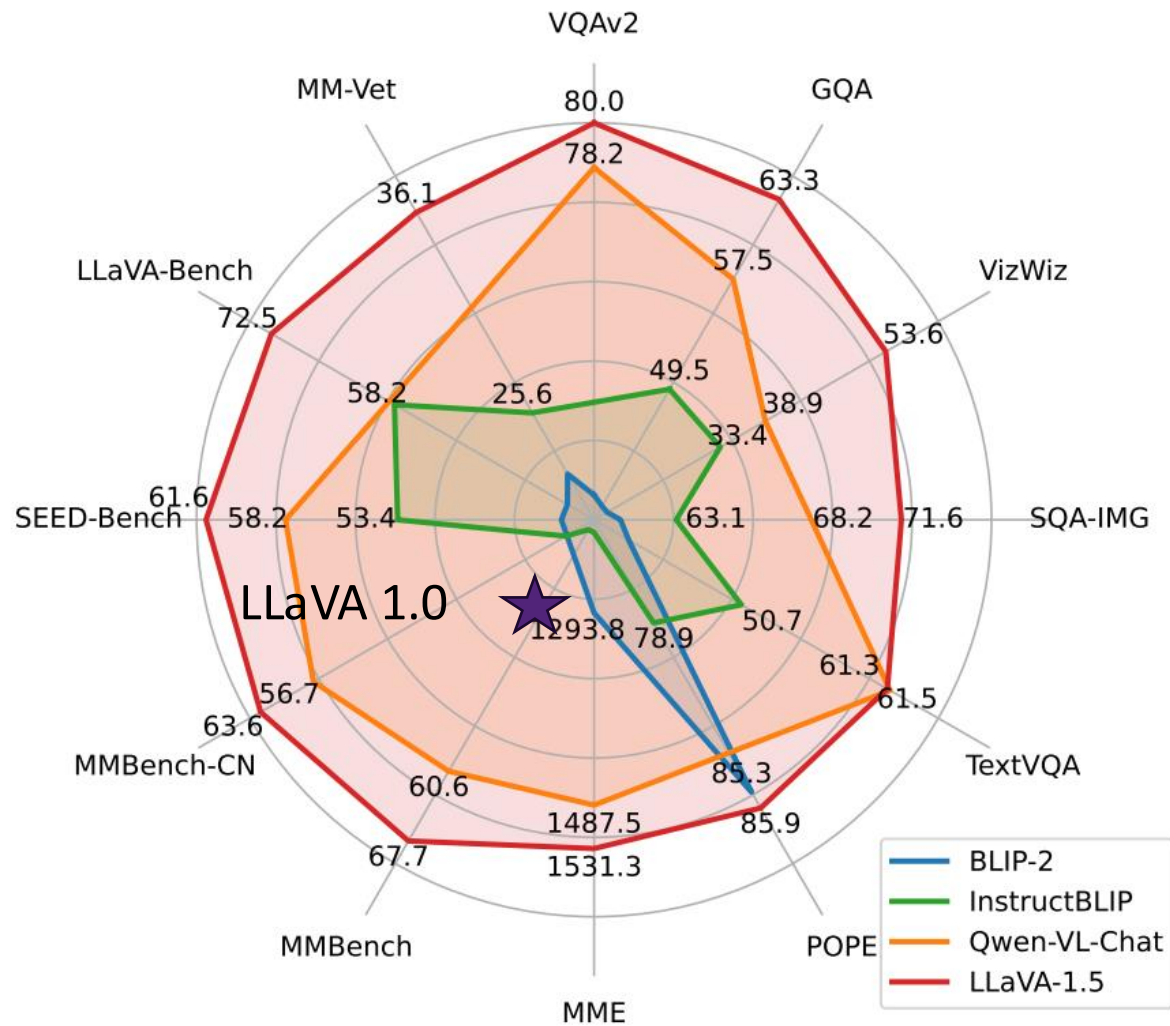


VQA-v2 Example



- A1. Is the **tray** on top of the **table** black or light brown? light brown
 A2. Are the **napkin** and the **cup** the same color? yes
 A3. Is the small **table** both oval and wooden? yes
 A4. Is there any **fruit** to the left of the **tray** the **cup** is on top of? yes
 A5. Are there any **cups** to the left of the **tray** on top of the **table**? no
 B1. What is the brown **animal** sitting inside of? **box**
 B2. What is the large **container** made of? cardboard
 B3. What **animal** is in the **box**? **bear**
 B4. Is there a **bag** to the right of the green **door**? no
 B5. Is there a **box** inside the plastic **bag**? no

LLaVA 1.5: More Data



Method	LLM	Res.	GQA	MME	MM-Vet
InstructBLIP	14B	224	49.5	1212.8	25.6

Only using a subset of InstructBLIP training data

0	LLaVA	7B	224	—	809.6	25.5
1	+VQA-v2	7B	224	47.0	1197.0	27.7
2	+Format prompt	7B	224	46.8	1323.8	26.3
3	+MLP VL connector	7B	224	47.3	1355.2	27.8
4	+OKVQA/OCR	7B	224	50.0	1377.6	29.6

Additional scaling

5	+Region-level VQA	7B	224	50.3	1426.5	30.8
6	+Scale up resolution	7B	336	51.4	1450	30.3
7	+GQA	7B	336	62.0*	1469.2	30.7
8	+ShareGPT	7B	336	62.0*	1510.7	31.1
9	+Scale up LLM	13B	336	63.3*	1531.3	36.1

Expanded from 158K to **665K** samples...

New Bottleneck: The Resolution Compute Trap

Visual input example, Constrained JSON Output:



User Please read the text in this image and return the information in the following JSON format (note xxx is placeholder, if the information is not available in the image, put "N/A" instead).

```
{"class": xxx, "DLN": xxx, "DOB": xxx, "Name": xxx, "Address": xxx, "EXP": xxx, "ISS": xxx, "SEX": xxx, "HGT": xxx, "WGT": xxx, "EYES": xxx, "HAIR": xxx, "DONOR": xxx}
```

GPT-4V {"class": "C", "DLN": "1234568", "DOB": "08/31/1977", "Name": "Ima Cardholder", "Address": "2570 24th Street, Anytown, CA 95818", "EXP": "08/31/2014", "ISS": "08/31/2011", "SEX": "F", "HGT": "5-05", "WGT": "125 lbs", "EYES": "BRN", "HAIR": "BRN", "DONOR": "VETERAN"}

LLaVA {"class": "Driver License", "DLN": "CA123 456 789", "DOB": "01/01/1990", "Name": "Amy", "Address": "Anytown, CA", "EXP": "08/31/2023", "ISS": "Veteran", "SEX": "F", "HGT": "5'5", "WGT": "130", "EYES": "Brown", "HAIR": "Brown", "DONOR": "Unknown"}

LLaVA-1.5 {"class": "C", "DLN": "1132456789", "DOB": "08/23/1971", "Name": "Ima Cardholder", "Address": "Anytown, State, Zip", "EXP": "08/23/2014", "ISS": "California", "SEX": "F", "HGT": "5'5", "WGT": "123", "EYES": "Brown", "HAIR": "Brown", "DONOR": "N/A"}

Still plenty of hallucinations...

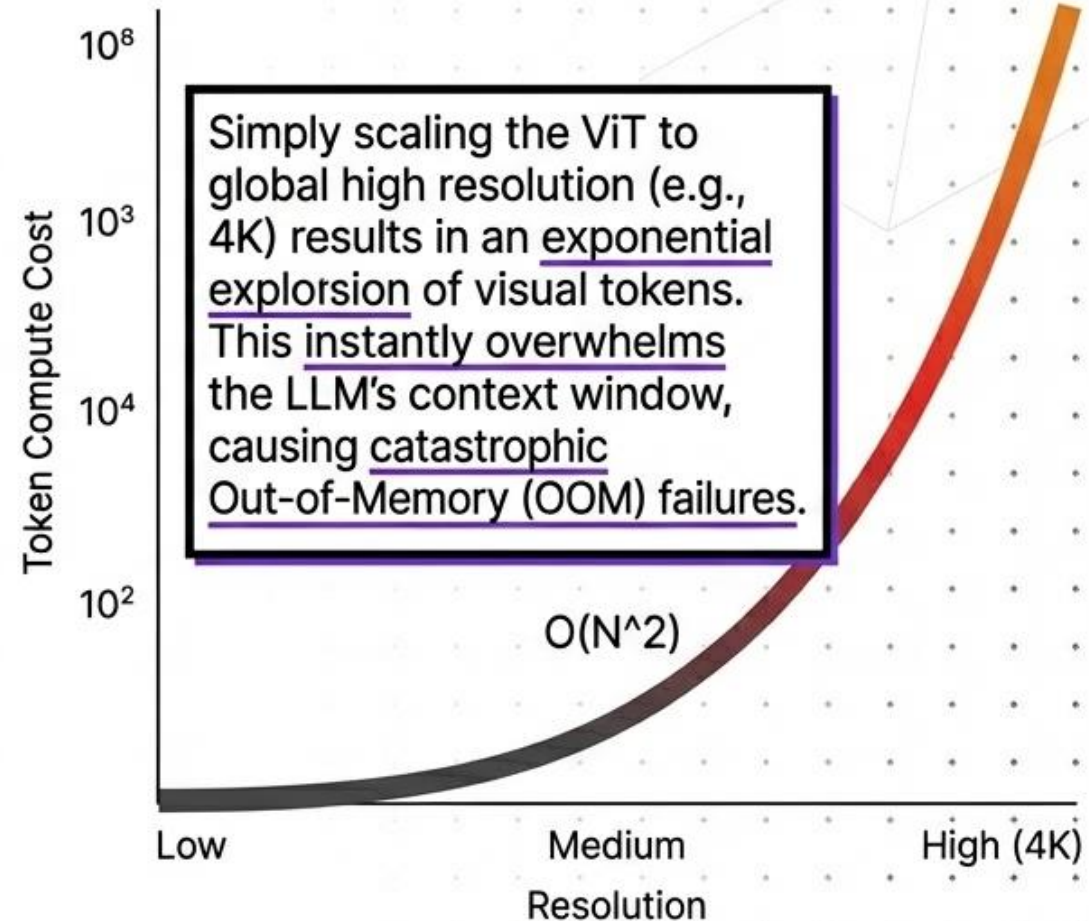
New Bottleneck: The Resolution Compute Trap

The 336 x 336 Limit



At **336px**, dense documents, complex charts, and high-res webpages turn into illegible blurs. The model is effectively blind to macro-documents.

The OOM Explosion



LLaVA-1.5-HD

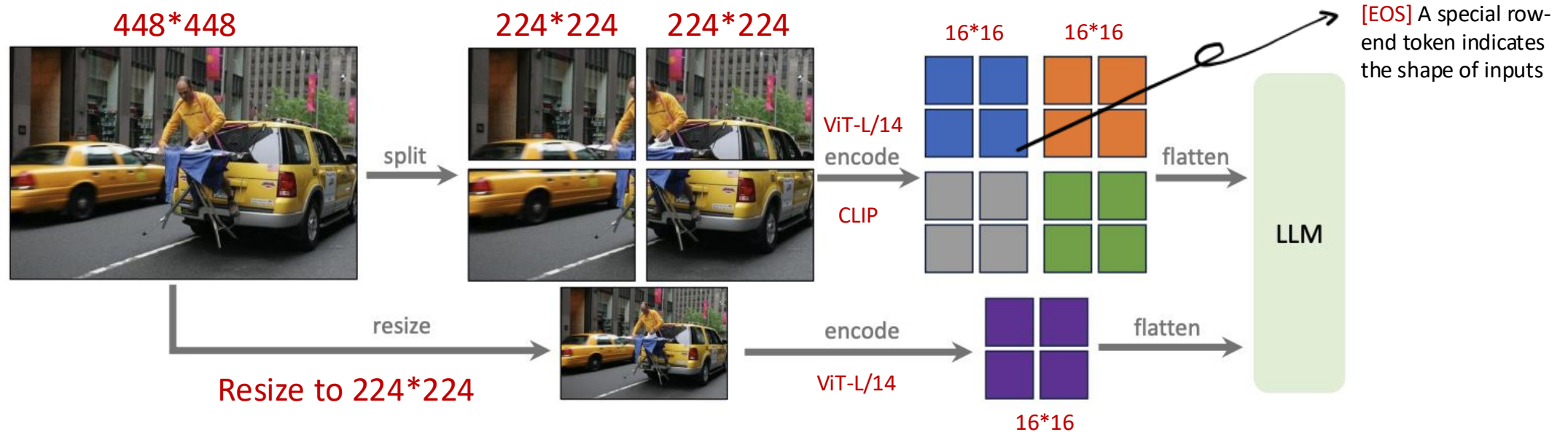


Figure 2. **LLaVA-1.5-HD**. Scaling LLaVA-1.5 to higher resolutions by splitting the image into grids and encoding them independently. This allows the model to scale to any resolution, without performing positional embedding interpolation for ViTs. We additionally concatenate the feature of a downsampled image to provide the LLM with a global context.

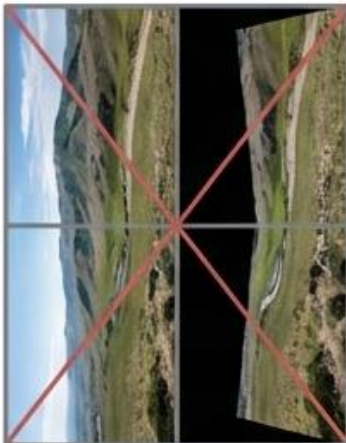
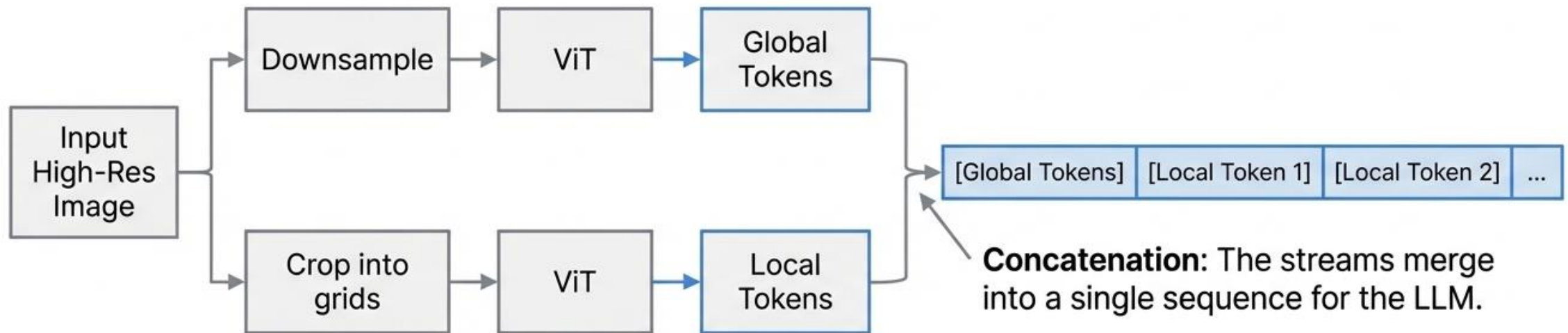
- do **not** perform additional high resolution pretraining for the visual projectors

LlaVA-1.5-HD

Method	LLM	Image Size	Sample Size		VQAv2 [19]	GQA [21]	VisWiz [20]	SciQA-IMG [38]	TextVQA [48]
			Pretrain	Finetune					
BLIP-2 [32]	Vicuna-13B	224 ²	129M	-	65.0	41	19.6	61	42.5
InstructBLIP [14]	Vicuna-7B	224 ²	129M	1.2M	–	49.2	34.5	60.5	50.1
InstructBLIP [14]	Vicuna-13B	224 ²	129M	1.2M	–	49.5	33.4	63.1	50.7
Shikra [8]	Vicuna-13B	224 ²	600K	5.5M	77.4*	–	–	–	–
IDEFICS-9B [22]	LLaMA-7B	224 ²	353M	1M	50.9	38.4	35.5	–	25.9
IDEFICS-80B [22]	LLaMA-65B	224 ²	353M	1M	60.0	45.2	36.0	–	30.9
Qwen-VL [3]	Qwen-7B	448 ²	1.4B [†]	50M [†]	<u>78.8*</u>	59.3*	35.2	67.1	63.8*
Qwen-VL-Chat [3]	Qwen-7B	448 ²	1.4B*	50M [†]	78.2*	57.5*	38.9	68.2	<u>61.5*</u>
LLaVA-1.5	Vicuna-7B	336 ²	558K	665K	<u>78.5*</u>	<u>62.0*</u>	<u>50.0</u>	66.8	58.2
LLaVA-1.5	Vicuna-13B	336 ²	558K	665K	80.0*	63.3*	53.6	71.6	<u>61.3</u>
LLaVA-1.5-HD	Vicuna-13B	448 ²	558K	665K	81.8*	64.7*	57.5	<u>71.0</u>	<u>62.5</u>
Specialist SOTA: PaLI-X-55B [11]					86.1*	72.1*	70.9*	–	71.4*

- Scaling up the resolution can effectively alleviate the hallucination issue...

Higher resolution?



The Limitation: LLaVA-HD relies on a hard-coded grid split (e.g., strictly 2x2 or 3x3). It fails to elegantly handle highly distorted aspect ratios, stretching or cropping vital data.

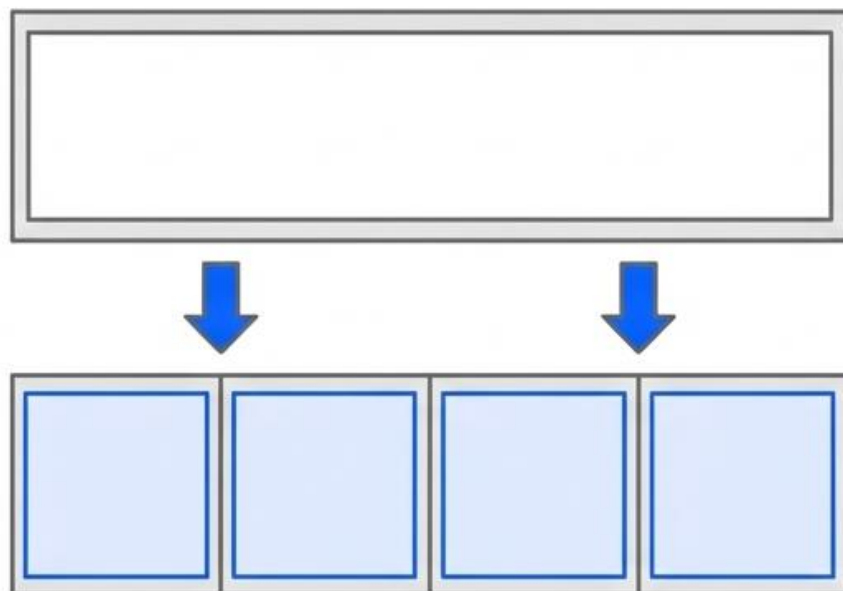
Damage the task performance of OCR, small object understanding...

LlaVA-1.6 (LlaVA-Next)

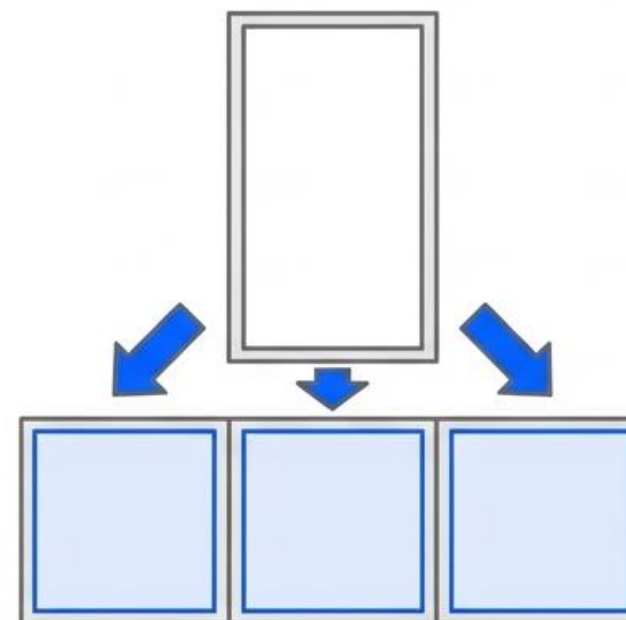
Preserve native aspect ratio:

336 x [(2,2), (1,2), (2,1), (1,3), (3,1), (1,4), (4,1)]

The Birth of AnyRes: To support wild, real-world images—like ultra-wide panoramic banners or long, scrolling documents—resolution slicing must become dynamically calculated.



Panoramic Banner -> 1x4 Grid



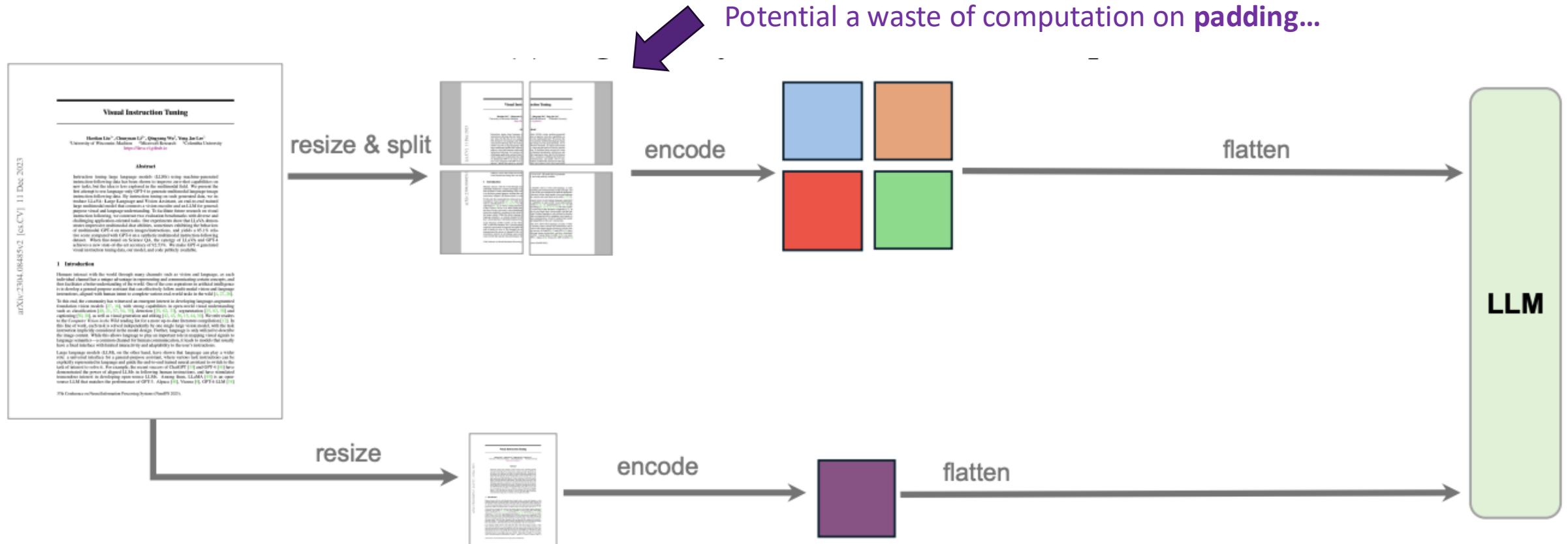
Vertical Document -> 3x1 Grid

The Core Principle: Use the minimum number of base 336x336 grids to construct the shape of the original image. This scales perception without stretching the image or altering its native aspect ratio.

LlaVA-1.6 (LlaVA-Next)

Preserve native aspect ratio:

336 x [(2,2), (1,2), (2,1), (1,3), (3,1), (1,4), (4,1)]

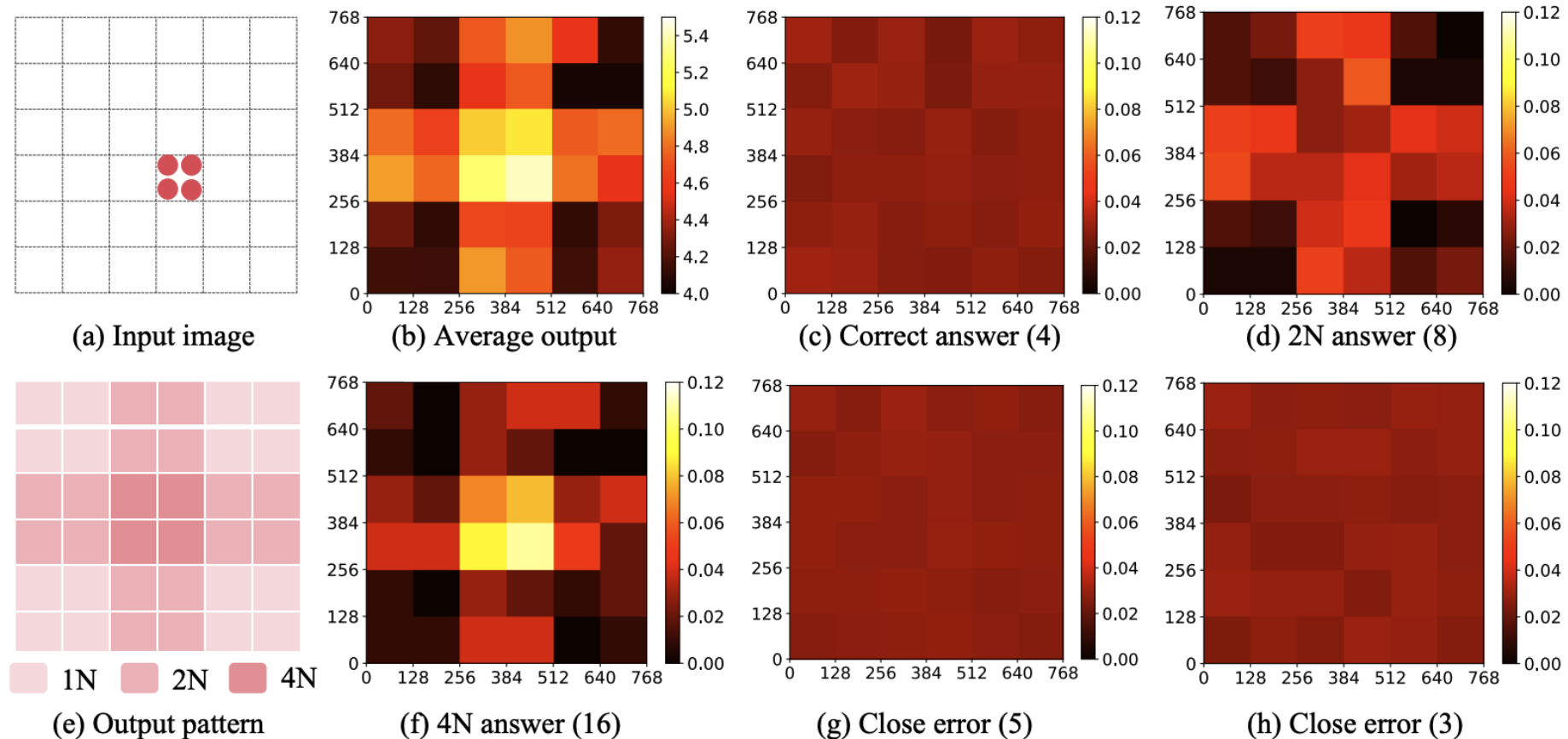


Open Question: MLLM needs higher resolution

AnyRes is a promising solution... While it is more important for us to figure out which step leads the model to generate the **wrong** answer

ask GPT-4V: "How many circles are there in the image?"

- (1) **Center:**
most
overestimate
d
- (2) **Middle**
Edge: less
overestimate
d
- (3) **Corner:**
closest to
correct



Why?

Open Question: MLLM needs higher resolution

Model responses show a significant **phase change** with image resolutions

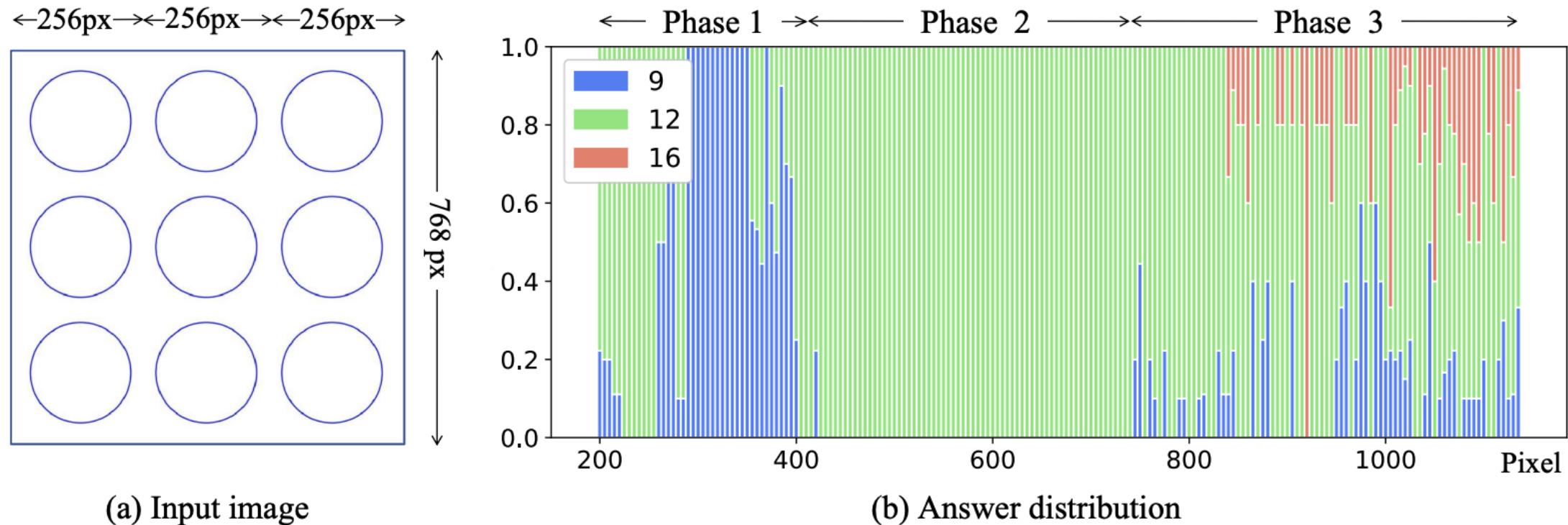
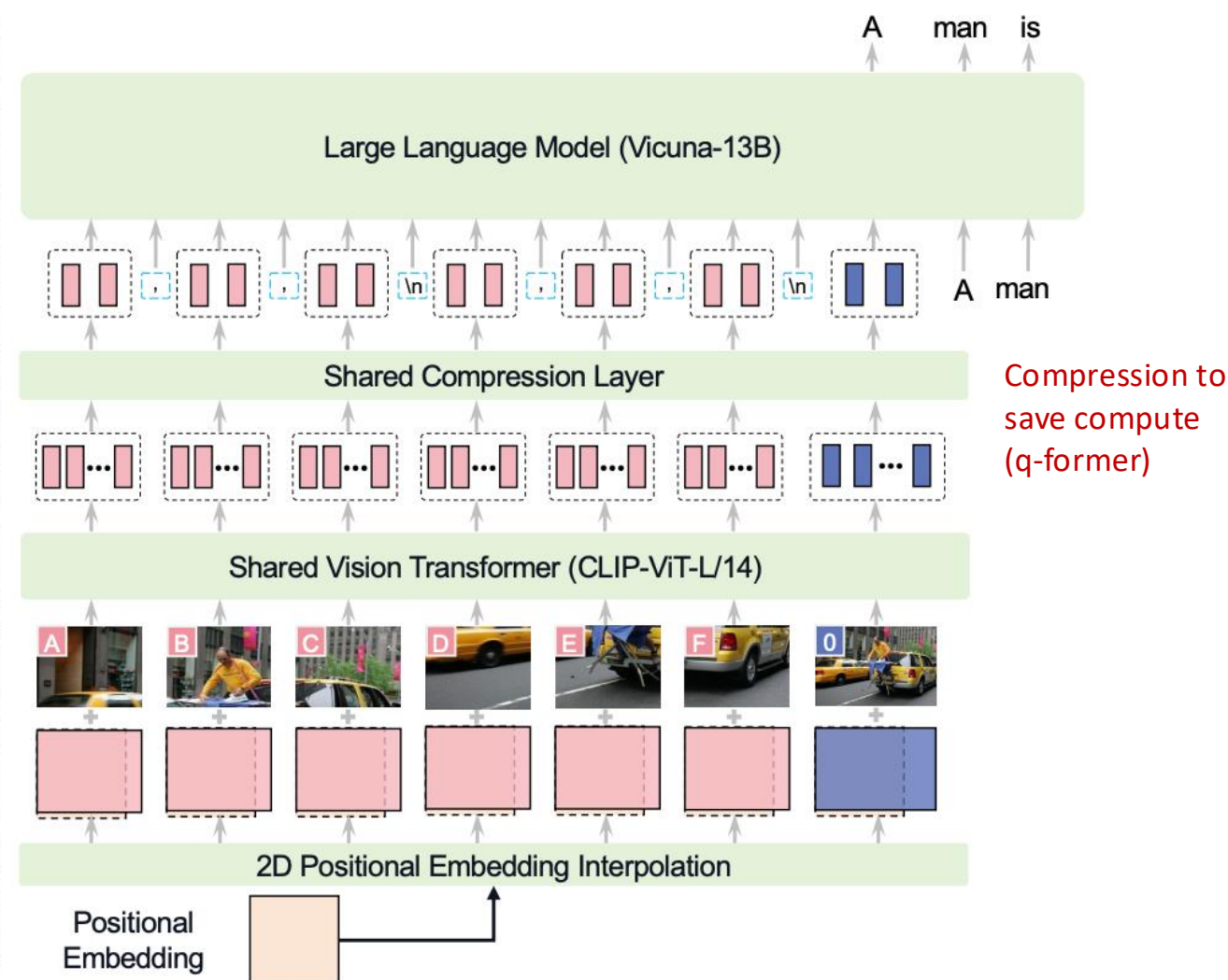
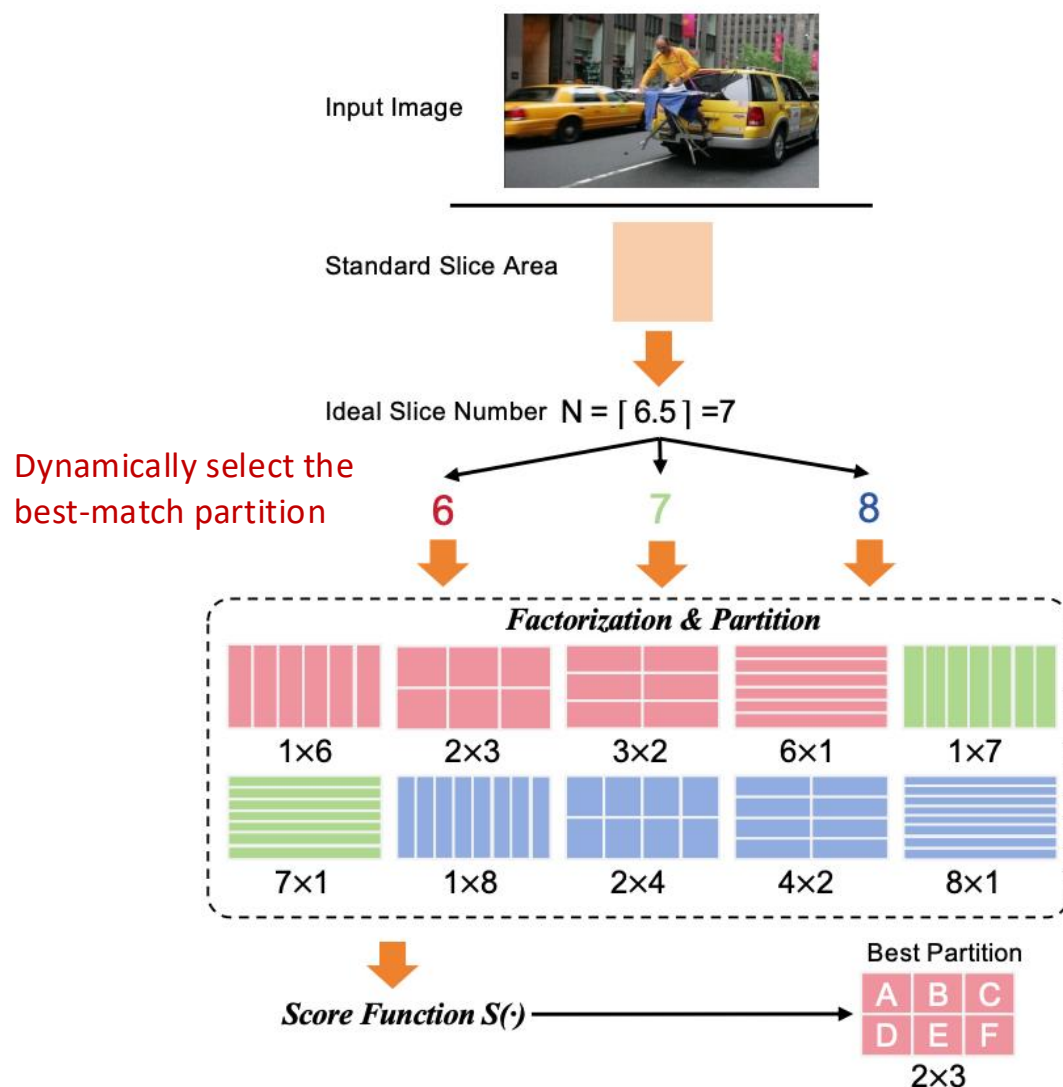


Figure 2: Results on probing GPT-4V via continuously changing image resolutions.

Open Question: MLLM needs higher resolution



Open Question: MLLM needs higher resolution

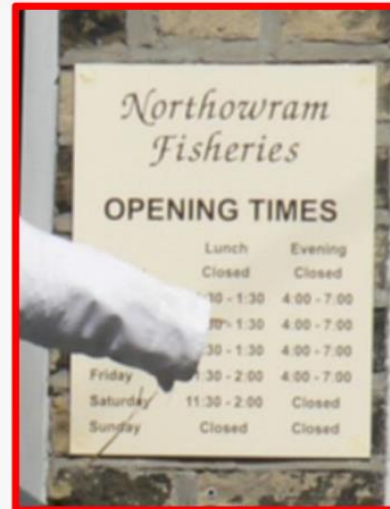
Model	#Data	MaxRes.	AR.	TFLOPs	VQA ^{v2}	GQA	VQA ^T	POPE	SQA	VizWiz	MME	MMB	MMB ^{CN}
BLIP-2 [21]	129M	224×224	Fix	1.0	41.0	41.0	42.5	85.3	61.0	19.6	1293.8	-	-
InstructBLIP [11]	130M	224×224	Fix	1.0	-	49.5	50.7	78.9	63.1	33.4	1212.8	-	-
Shikra [8]	6M	224×224	Fix	8.0	77.4	-	-	-	-	-	-	58.8	-
Qwen-VL [5]	1.4B	448×448	Fix	9.2	78.8	59.3	<u>63.8</u>	-	67.1	35.2	-	38.2	7.4
SPHINX [24]	1.0B	448×448	Fix	39.7	78.1	62.6	51.6	80.7	69.3	39.9	1476.1	66.9	56.2
SPHINX-2k [24]	1.0B	762×762	Fix	69.4	<u>80.7</u>	63.1	61.2	<u>87.2</u>	70.6	44.9	1470.7	65.9	57.9
MiniGPT-v2 [7]	326M	448×448	Fix	4.3	-	60.1	-	-	-	53.6	-	-	-
Fuyu-8B [6]	-	1024×1024	Any	21.3	74.2	-	-	74.1	-	-	728.6	10.7	-
OtterHD-8B [20]	-	1024×1024	Any	21.3	-	-	-	86.0	-	-	1223.4	58.3	-
mPLUG-Owl2 [43]	401M	448×448	Fix	1.7	79.4	56.1	58.2	86.2	68.7	54.5	1450.2	64.5	-
UReader [42]	86M	896×1120	Enum	26.0	-	-	57.6	-	-	-	-	-	-
Monkey [23]	1.0B	896×1344	Enum	65.3	80.3	60.7	-	67.6	69.4	61.2	-	-	-
LLaVA-1.5 [27]	1.2M	336×336	Fix	15.5	80.0	<u>63.3</u>	61.3	85.9	<u>71.6</u>	53.6	<u>1531.3</u>	<u>67.7</u>	<u>63.6</u>
LLaVA-UHD (ours)	1.2M	672×1008	Any	14.6	81.7	65.2	67.7	89.1	72.0	<u>56.1</u>	1535.0	68.0	64.8
Δ	-	×6 times	-	-0.9	+1.7	+1.9	+6.4	+3.2	+0.4	+2.5	+3.7	+0.3	+1.2

OCR capacities improve significantly without new data

Open Question: MLLM needs higher resolution



What is the opening time of this restaurant ?



LLaVA-UHD (Ours)

The image shows a sign with the opening times for the restaurant. It appears to be open from **11:30AM to 1:30 PM and from 4:00 PM to 7:00 PM from Monday to Saturday**. The restaurant is **closed on Sundays**.



LLaVA-1.5

The image shows a sign with the opening times for the restaurant. It appears to be open from **12:00 to 2:00 PM and from 5:00 PM to 7:00 PM**.



What is written on the poster ?



LLaVA-UHD (Ours)

The poster in the image displays the text "**SAVER MENU**." This suggests that the restaurant is advertising a menu with lower-priced items, which is a common marketing strategy for fast-food restaurants to attract customers looking for affordable options.



LLaVA-1.5

The poster in the image shows **a menu with various food items**, but the text is **too small to read clearly**. It appears to be a promotional poster for the McDonald's restaurant, likely displaying their menu offerings or special deals.

Open Question: MLLM needs higher resolution?

- Beyond up-resizing, feeding multiple crops... **Proactively searching for RoI**

Question:
What is the number on the left brown bus?

LLaVA-1.5:
The number on the left brown bus is **10**. 🤔

LLaVA-1.5 - CoS:
Ans 1: The region of interest is:

Ans 2: The number is **95**. 😊

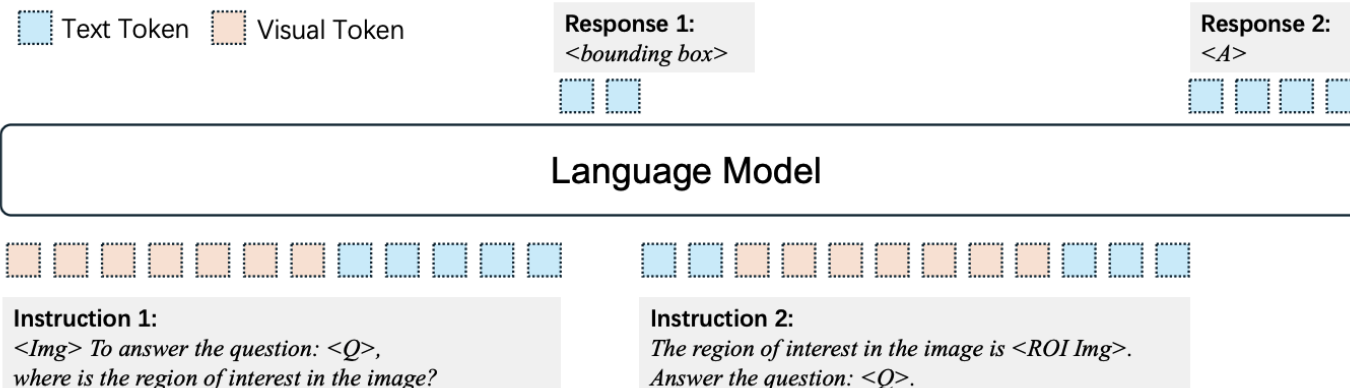


Question:
What color are the two cars in the middle front?

LLaVA-1.5:
The two cars in the middle front are **red**. 🤔

LLaVA-1.5 - CoS:
Ans 1: The region of interest is:

Ans 2: **Red and blue**. 😊



Open Question: MLLM needs higher resolution?

You are an advanced multimodal understanding assistant. You will be given an image and a question about it.

Provide Your Answer: Examining the image and the question thoroughly, answer the question in the format “**Answer: [Letter]**”, where [Letter] is only the letter (A, B, C, D, E) of the correct option.

Identify Critical Area: After output the answer, Please determine a bounding box (normalized coordinates, $0 \leq x1, y1, x2, y2 \leq 1$, $x1 < x2$, $y1 < y2$) of the area. Ensure the bounding box is large enough to include all the surrounding context that might be relevant to answering the question (such as information about the table row or column, nearby text, interacting objects, relevant background). Then output in the format “**Bounding Box: [x1, y1, x2, y2]**”.

I will provide you an extra image which is cropped from the original image. Just treat the newly input cropped image as an additional information to the local detail information. (The cropped image is clearer)

Please examine the cropped image. (The cropped image may not contain the information needed to answer the question, then ignore this cropped image)

Review the original image and combine with the information from the cropped image.

Identify potential omission of information in visual perception or calculation based on the image and question.

If you think your previous answer is **correct**, please reasoning to explain why and conclude your response with the format “**Rechecked Answer: [Letter]**”, where [Letter] is only the letter (A, B, C, D, E) of the correct option.

If you think your previous answer is **wrong**, please reasoning to explain why and correct the answer. Conclude your response with the format “**Rechecked Answer: [Letter]**”, where [Letter] is only the letter (A, B, C, D, E) of the correct option.

Self-refinement Prompt

- Examine the cropped image
- Review the original image and combine
- Identify possible omission
- Recheck answer

Force the MLLM to re-evaluate its initial answer using the nearly zoomed-in crop

Figure 2: Prompt engineering of Zoom-Refine. System prompt, "Localized Zoom" prompt, and "Self-refinement" prompt are highlighted in orange, blue, and green, respectively.

“structured prompting”

Open Question: MLLM needs higher resolution?

Table 1: Experimental results on the reasoning tasks of the MME-Realworld benchmark. OCR, DT, MO, and AD each indicate a specific subtask: Optical Character Recognition in the Wild, Diagram and Table, Monitoring, and Autonomous Driving, respectively. Avg and Avg-C indicate the weighted and unweighted average accuracy across subtasks in each subtask.

Method	OCR	DT	MO	AD	Avg	Avg-C
InternVL3-78B + Ours	75.40	71.80	47.39	40.22	53.22	58.70
InternVL3-78B [74]	72.80	62.20	43.78	38.47	49.62	54.31
InternVL2.5-78B + Ours	74.20	60.80	44.38	38.99	49.97	54.59
InternVL2.5-78B [7]	71.20	53.20	37.35	36.75	45.60	49.52
Qwen2-VL-7B [61]	63.40	48.60	33.13	31.47	40.39	44.15
InternVL2-7B [9]	57.40	39.00	42.57	30.84	38.74	42.45
GPT-4o [23]	61.40	44.00	42.57	30.84	37.61	42.28
CogVLM2-8B [17]	54.00	32.00	42.57	30.84	37.61	39.79
InternVL-20B [9]	56.80	35.40	37.35	28.94	30.48	39.62
Cambrian-1-8B [57]	53.20	27.40	42.37	30.73	36.16	38.43
SliME-8B [72]	53.20	29.40	36.14	31.55	35.80	37.57
MiniCPM-V2.5-8B [19]	44.00	31.80	36.95	31.03	34.50	35.95
SliME-13B [72]	41.00	39.00	33.13	30.80	34.46	35.98
InternLM-XComposer2.5-7B [71]	53.40	41.00	17.67	29.99	33.90	35.52
GPT-4o-mini [23]	47.00	39.80	25.81	26.79	32.48	34.85
YI-VL-34B [69]	42.40	26.00	31.33	31.55	32.45	32.82
LLaVA-Next-8B [29]	55.20	23.40	21.08	30.73	32.06	32.60
Mini-Gemini-34B-HD [35]	59.20	39.20	20.48	22.84	31.73	35.43
Gemini-1.5-pro [56]	52.70	33.20	28.33	19.20	29.19	33.36
Cambrian-1-34B [57]	55.00	36.00	19.48	16.07	27.06	31.64

Why does Qwen-VL show superior HR task performance?

Optical Character Recognition (OCR)



What is the speed limit on this road?

- (A) 70
- (B) 50**
- (C) 60
- (D) 80
- (E) The image does not feature the limitation

Monitoring (MO)

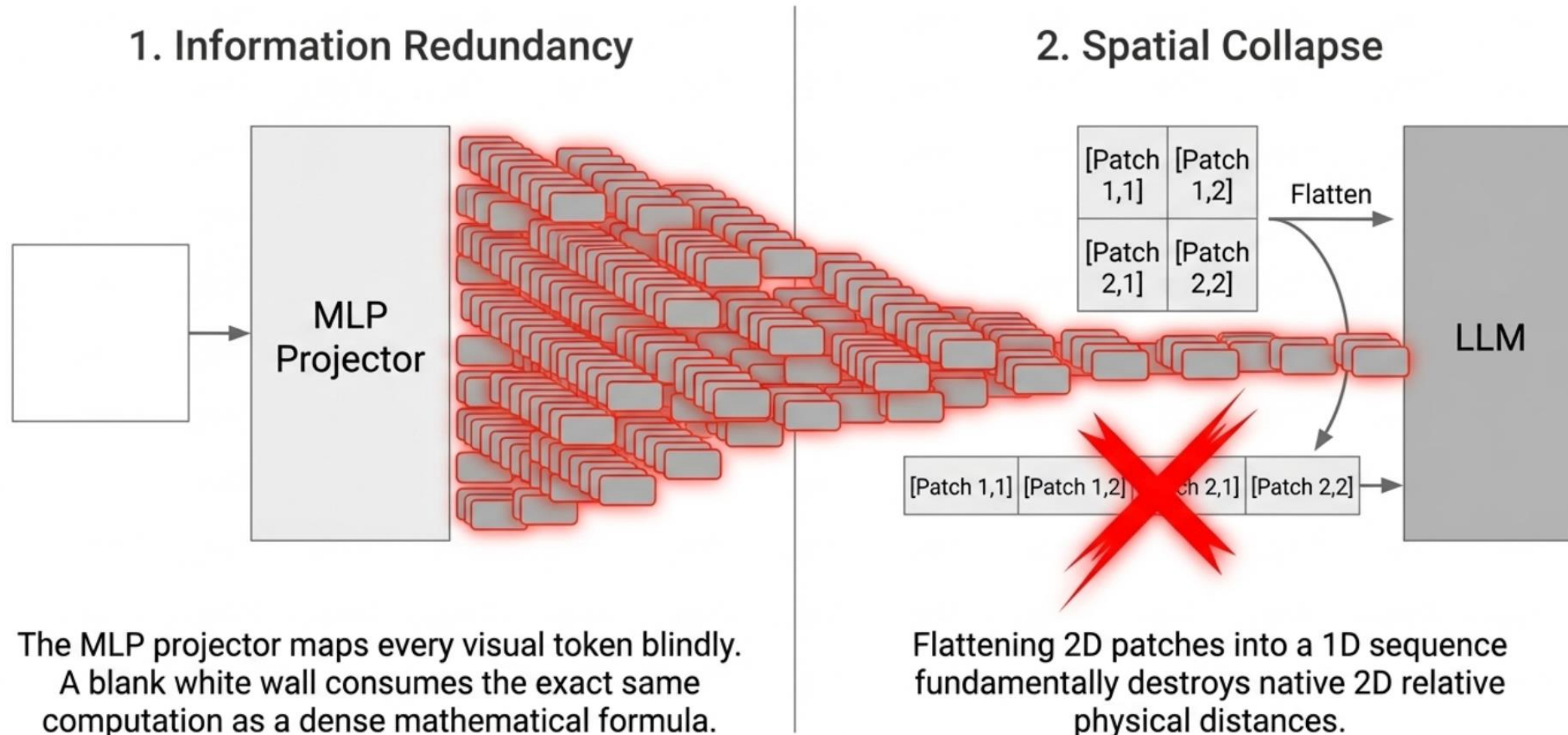


What's on the truck in the image?

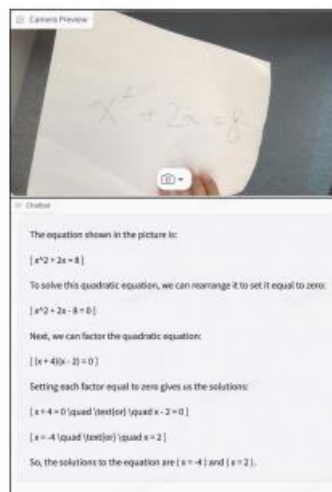
- (A) plastic
- (B) wood**
- (C) iron
- (D) water
- (E) The image does not feature the object**

Qwen2-VL

- LLaVA paradigm suffers from information redundancy and spatial collapse

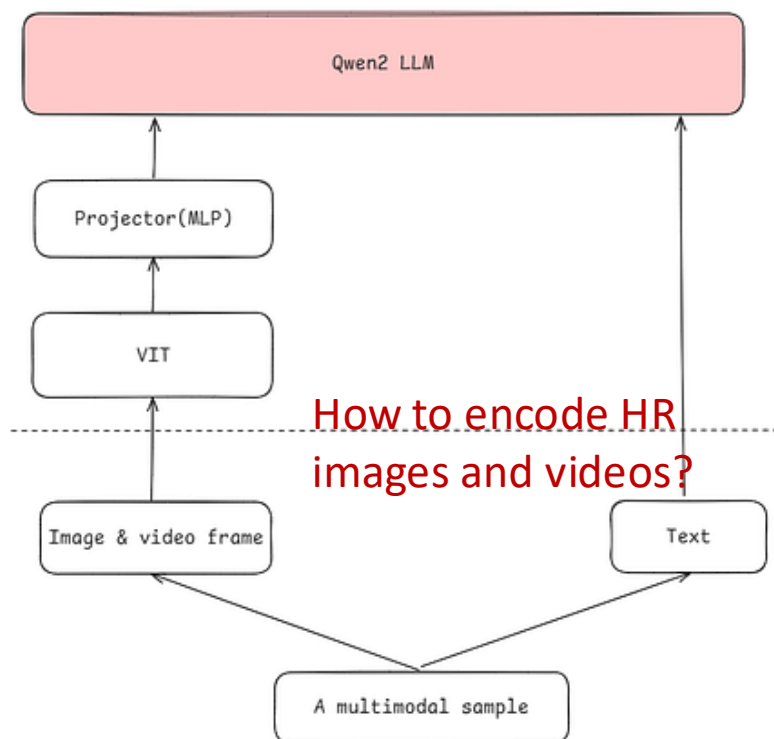


Qwen2-VL



Qwen2-VL

Qwen2-VL



Special token

The Dataset Format Example of ChatML

```

<|im_start|>user
<|vision_start|>Picture1.jpg<|vision_end|><|vision_start|>Picture2.jpg<|vision_end|>What do the
two pictures have in common?<|im_end|>
<|im_start|>assistant
Both pictures are of SpongeBob SquarePants. <|im_end|>
<|im_start|>user
What is happening in the video?<|vision_start|>video.mp4<|vision_end|><|im_end|>
<|im_start|>assistant
The protagonist in the video is frying an egg.<|im_end|>

```



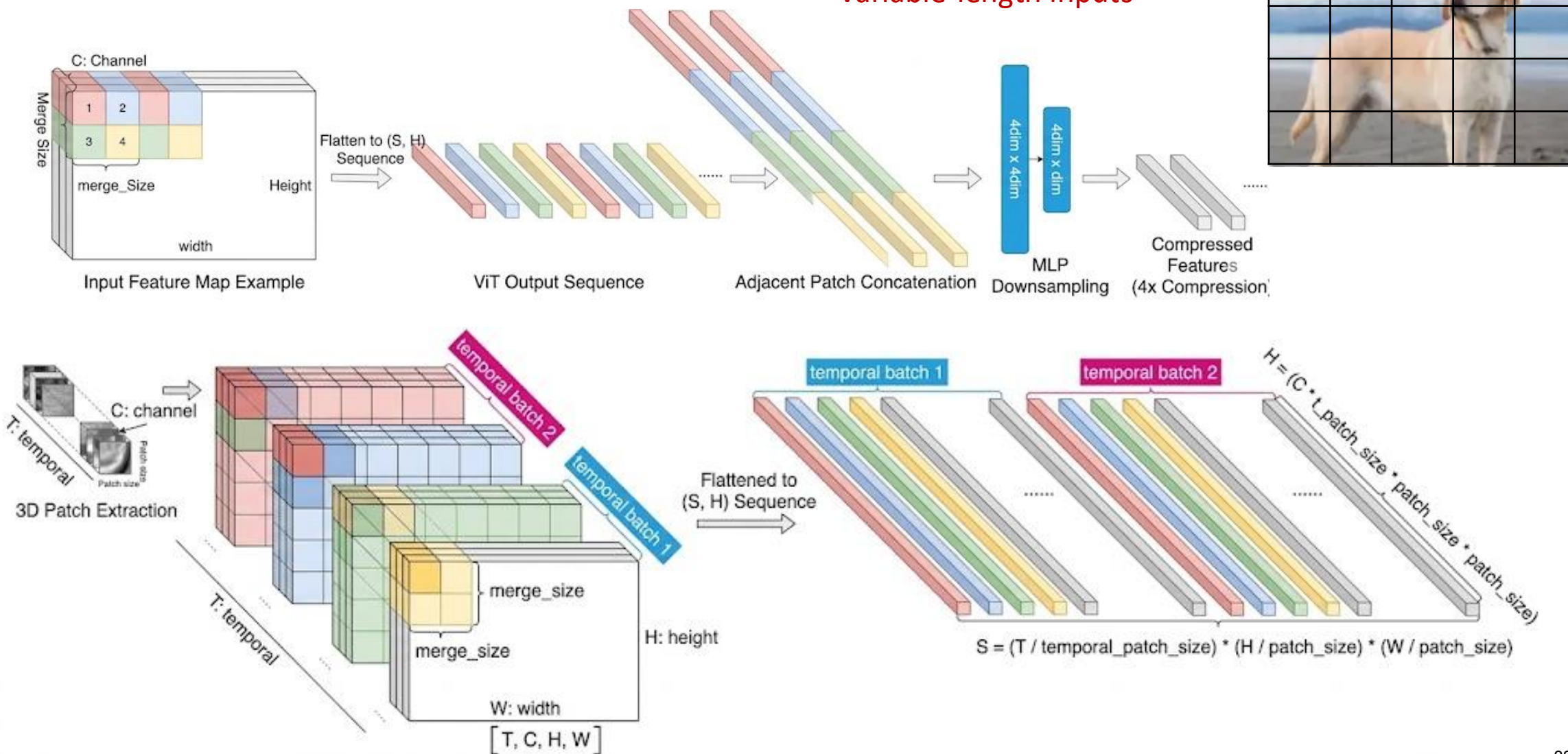
what do the two pictures have in common

The two pictures, while featuring different animals and settings, share several common themes and characteristics:

- **Subject Matter:** Both images feature a popular domestic pet. The first picture shows a dog (a yellow Labrador Retriever), and the second shows a cat (an orange tabby).
- **Portraiture Style:** Each image is a portrait focused on the animal. They are framed to highlight the creature's features against a softer background.

Qwen2-VL

Variable-length inputs

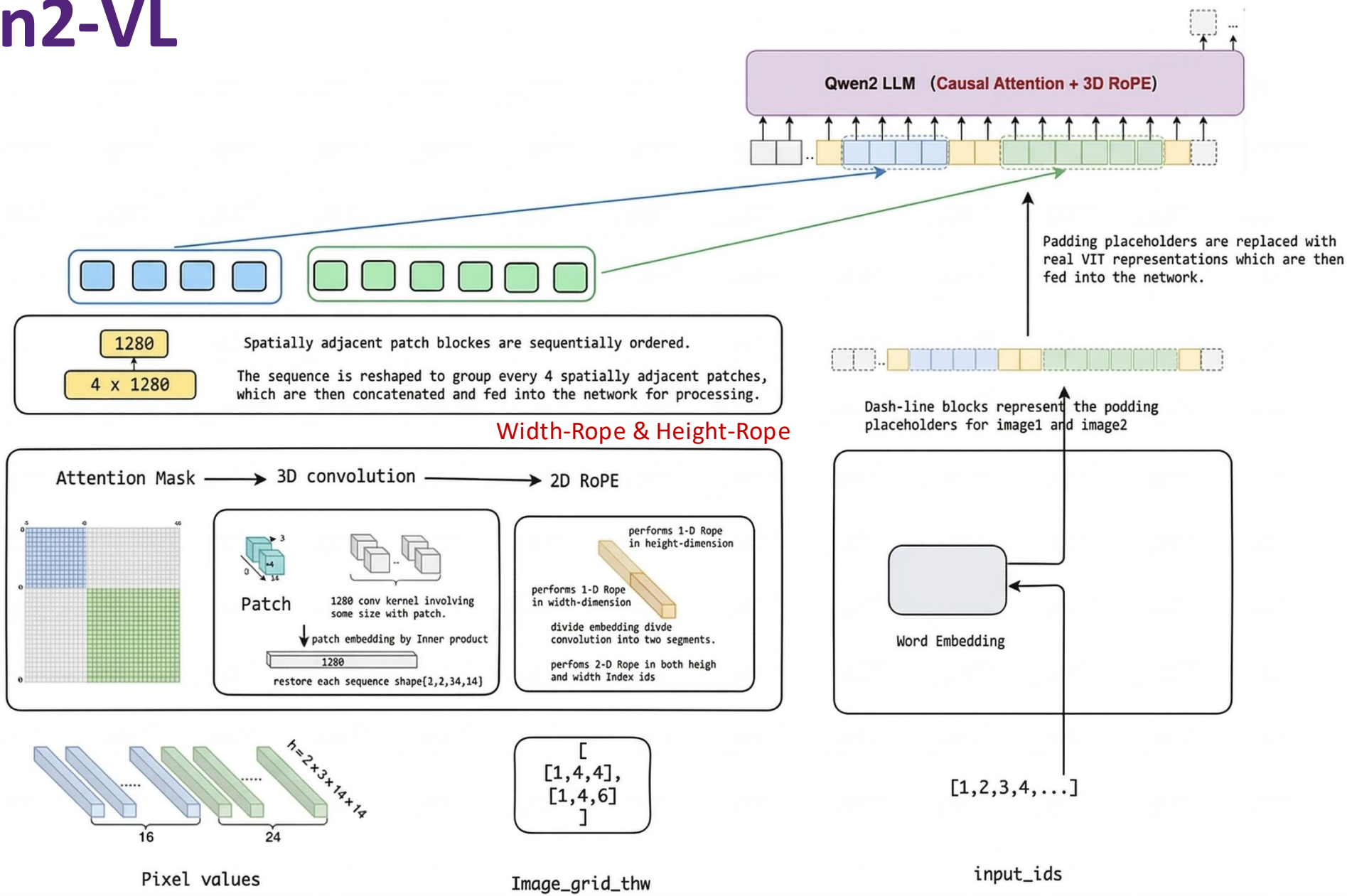


Qwen2-VL

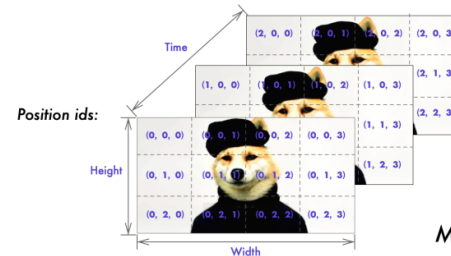
LLM

MLP

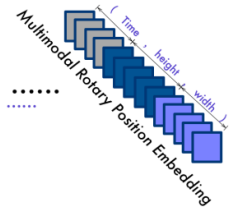
ViT



Qwen2-VL



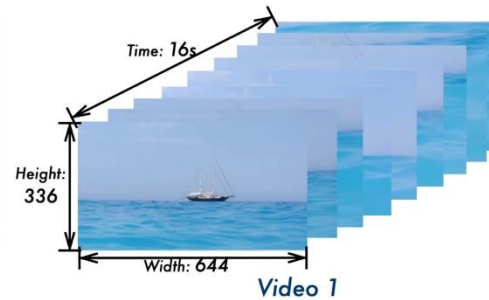
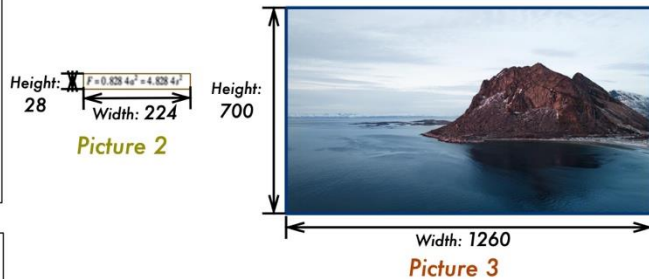
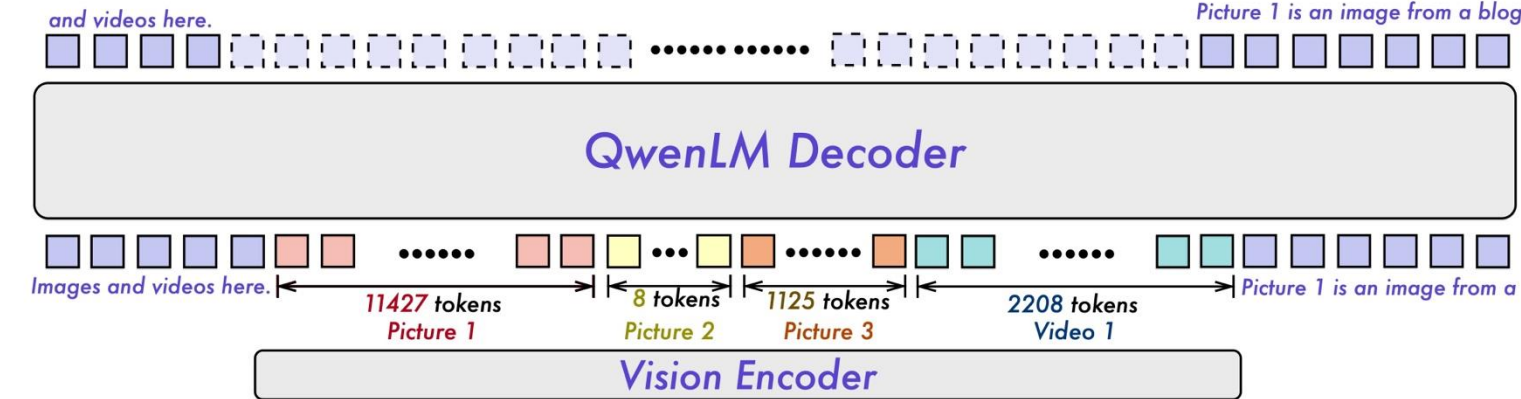
This video features a dog, specifically a Shiba
(4, 4, 4) (5, 5, 5) (6, 6, 6) (7, 7, 7) (8, 8, 8) (9, 9, 9) (10, 10, 10) (11, 11, 11) (12, 12, 12)



Multimodal Rotary Position Embedding (M-RoPE)

$$\mathcal{R}_{\Theta, m, n}^d = \begin{pmatrix} \text{X-axis rotation, } m & & & \\ \cos(m\theta) & -\sin(m\theta) & 0 & 0 \\ \sin(m\theta) & \cos(m\theta) & 0 & 0 \\ 0 & 0 & \text{Y-axis rotation, } n & \\ 0 & 0 & \cos(n\theta) & -\sin(n\theta) \\ & & \sin(n\theta) & \cos(n\theta) \end{pmatrix}$$

Enable the model to extrapolate to longer sequences during inference



Qwen2-VL

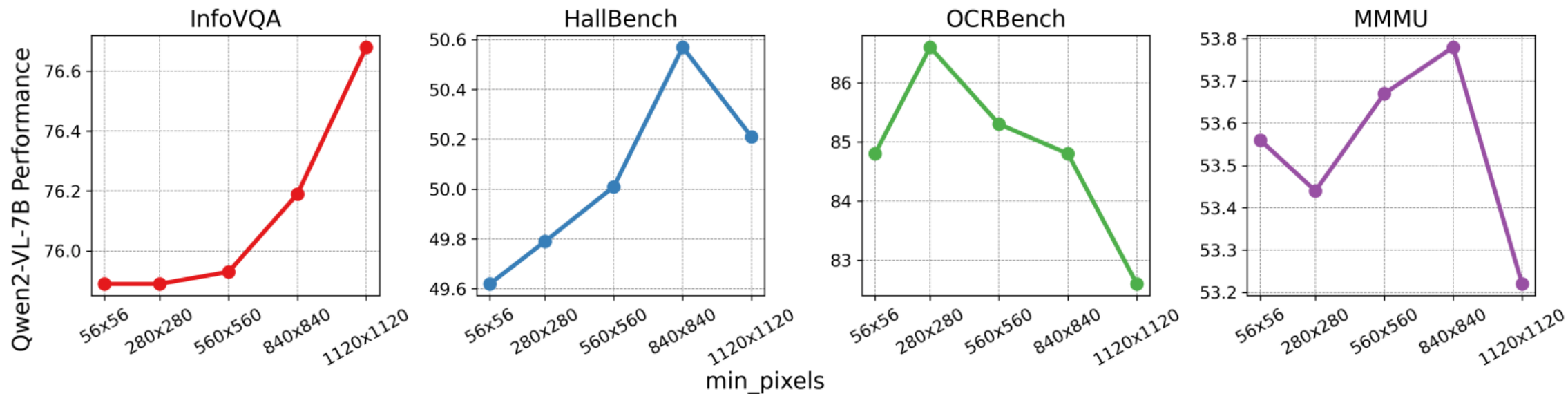
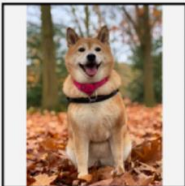
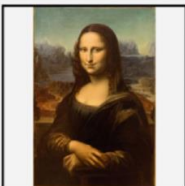
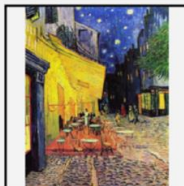
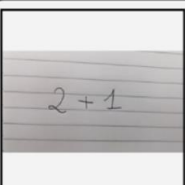
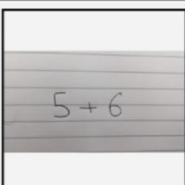
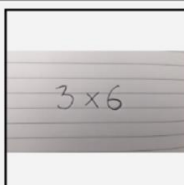


	Image Benchmarks								Video Benchmarks		
	MathVista	MMB	MMStar	RWQ	DocVQA	ChartQA	InfoVQA	TextVQA	PerceptionTest	NextQA	STAR
1D-RoPE	39.2	58.6	36.7	54.5	82.5	68.0	50.8	71.3	46.6	43.9	55.5
M-RoPE	43.4	60.6	36.7	53.7	82.8	68.4	50.3	71.8	47.4	46.0	57.9

Qwen2-1.5B

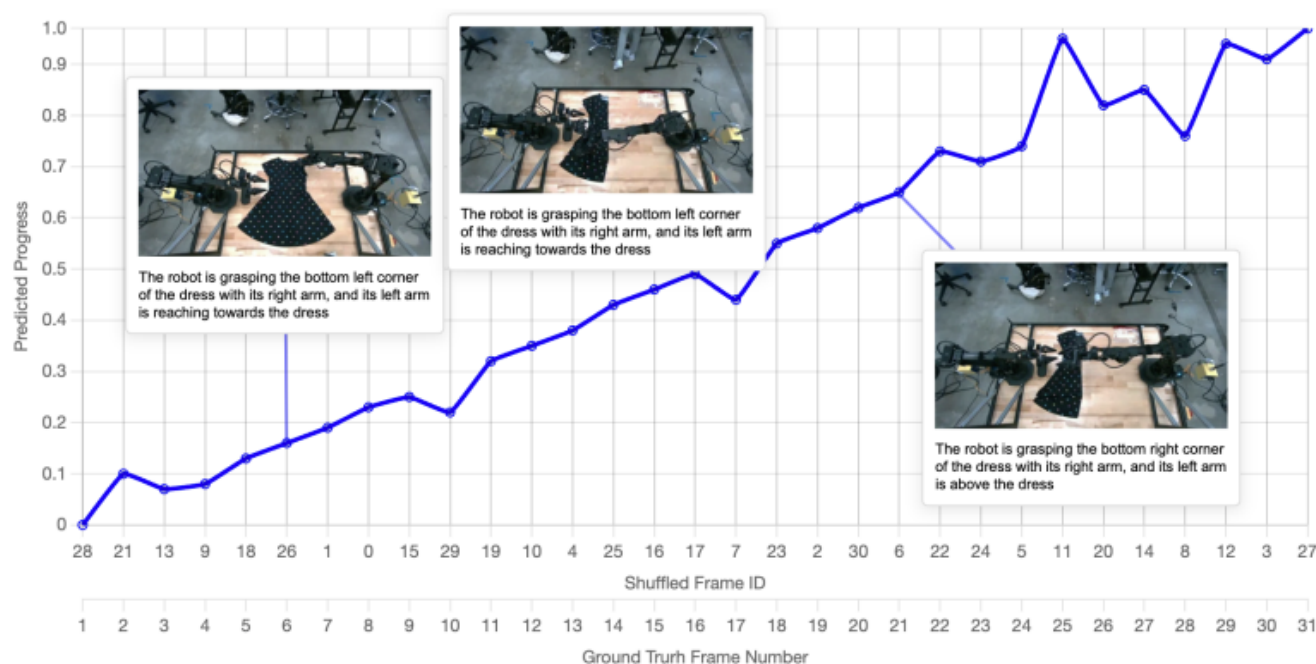
Multimodal In-context Learning

- the ability of a model to infer a task from a few interleaved examples of inputs and outputs across multiple modalities, **without** updating its parameters.

Input Prompt					Completion	
	This is a chinchilla. They are mainly found in Chile.		This is a shiba. They are very popular in Japan.		This is	→ a flamingo. They are found in the Caribbean and South America.
	What is the title of this painting? Answer: The Hallucinogenic Toreador.		Where is this painting displayed? Answer: Louvres Museum, Paris.		What is the name of the city where this was painted? Answer:	→ Arles.
	Output: "Underground"		Output: "Congress"		Output:	→ "Soulomes"
	2+1=3		5+6=11			→ 3x6=18

Open Question: Multimodal In-context Learning

- Why is multimodal ICL important?



(a) `fold_dress` from the top-down view.

Google DeepMind

2024-11-8

Vision Language Models are In-Context Value Learners

Yecheng Jason Ma^{†,1,2}, Joey Hejna^{1,3}, Ayzaan Wahid¹, Chuyuan Fu¹, Dhruv Shah¹, Jacky Liang¹, Zhuo Xu¹, Sean Kirmani¹, Peng Xu¹, Danny Driess¹, Ted Xiao¹, Jonathan Tompson¹, Osbert Bastani², Dinesh Jayaraman², Wenhao Yu¹, Tingnan Zhang¹, Dorsa Sadigh^{1,3}, Fei Xia¹

¹Google DeepMind, ²University of Pennsylvania, ³Stanford University

Correspond to: jasonyma@seas.upenn.edu, xiafei@google.com

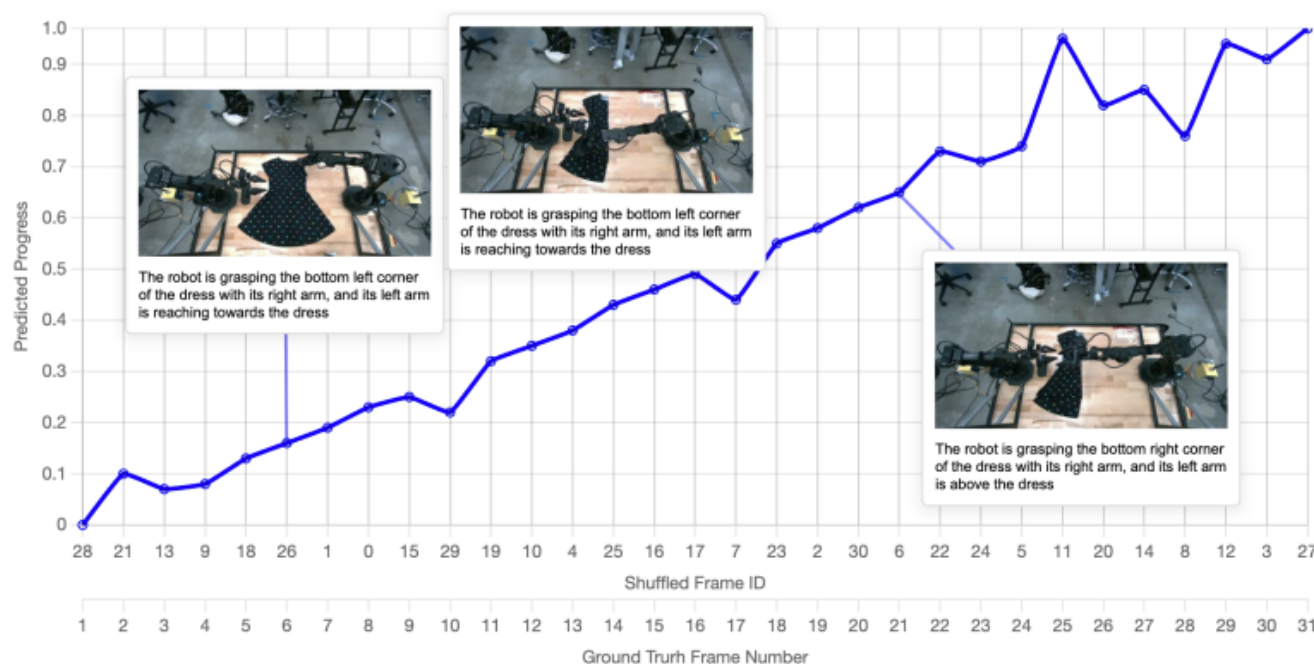
Website and Interactive Demo: generative-value-learning.github.io

Abstract

Predicting temporal progress from visual trajectories is important for intelligent robots that can learn, adapt, and improve. However, learning such progress estimator, or temporal value function, across different tasks and domains requires both a large amount of diverse data and methods which can scale and generalize. To address these challenges, we present Generative Value Learning (GVL), a universal value function estimator that leverages the world knowledge embedded in vision-language models (VLMs) to predict task progress. Naively asking a VLM to predict values for a video sequence performs poorly due to the strong temporal correlation between successive frames. Instead, GVL poses value estimation as a temporal ordering problem over shuffled video frames; this seemingly more challenging task encourages VLMs to more fully exploit their underlying semantic and temporal grounding capabilities to differentiate frames based on their perceived task progress, consequently producing significantly better value predictions. Without any robot or task specific training, GVL can in-context zero-shot and few-shot predict effective values for more than 300 distinct real-world tasks across diverse robot platforms, including challenging bimanual manipulation tasks. Furthermore, we demonstrate that GVL permits flexible multi-modal in-context learning via examples from heterogeneous tasks and embodiments, such as human videos. The generality of GVL enables various downstream applications pertinent to visuomotor policy learning, including dataset filtering, success detection, and advantage-weighted regression – all without any model training or finetuning.

Open Question: Multimodal In-context Learning

- Why is multimodal ICL important?



(a) fold_dress from the top-down view.

You are an expert roboticist tasked to predict task completion percentages for frames of a robot for the task of {task_description}. The task completion percentages are between 0 and 100, where 100 corresponds to full task completion. We provide several examples of the robot performing the task at various stages and their corresponding task completion percentages. Note that these frames are in random order, so please pay attention to the individual frames when reasoning about task completion percentage.

Initial robot scene: [IMG]

In the initial robot scene, the task completion percentage is 0.

Now, for the task of {task_description}, output the task completion percentage for the following frames that are presented in random order. For each frame, format your response as follow: Frame {i}: Frame Description: {}, Task Completion Percentages: {}%

Frame 1: [IMG]

...

Frame n: [IMG]

Quiz Time?



Open Question: Multimodal In-context Learning

- Why is multimodal ICL difficult?

In-context learning can be

sensitive to:

- The order of examples are presented
- The relevance of examples to the target question;
- ...

Question: Following the format of image-caption pairs, describe the final image.

 The man has big black eyes and short eyelashes.	 A man with a broad chin and a square face.	 The man has big black eyes and short eyelashes.
 The lady has a melon seed face and a narrow chin.	 The lady has a melon seed face and a narrow chin.	 A man with a broad chin and a square face.
 An old man with a broad forehead.	 An old man with a broad forehead.	 An old man with a broad forehead.
 A man with a broad chin and a square face.	 The man has big black eyes and short eyelashes.	 The lady has a melon seed face and a narrow chin.
 ?	 ?	 ?
		
OpenFlamingo	OpenFlamingo	OpenFlamingo
		
Answer: A black hair man with short eyelashes. 😭 ❌	Answer: A man with a broad chin. 😭 ❌	Answer: A young woman with a round face and a narrow chin. 😊 ✅

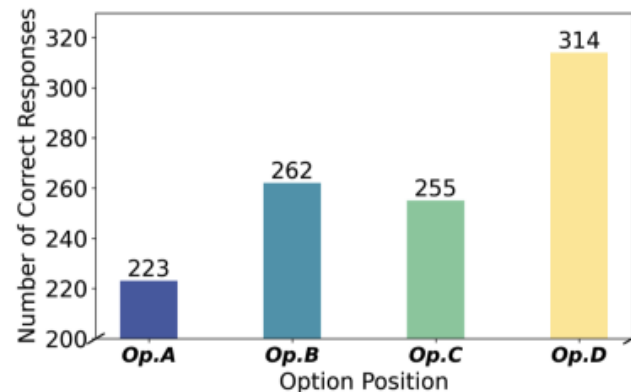
Open Question: Multimodal In-context Learning

- Why is multimodal ICL difficult?

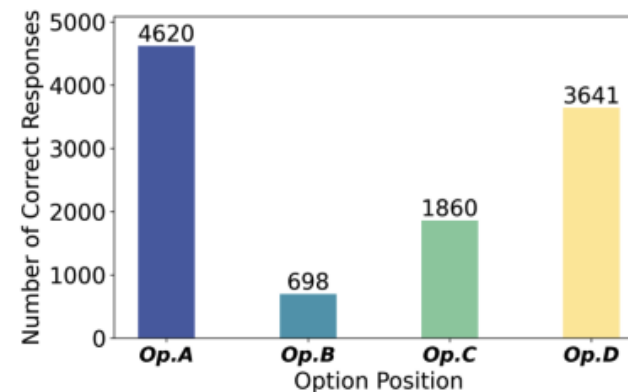
In-context learning can be **sensitive** to:

- The order of examples are presented

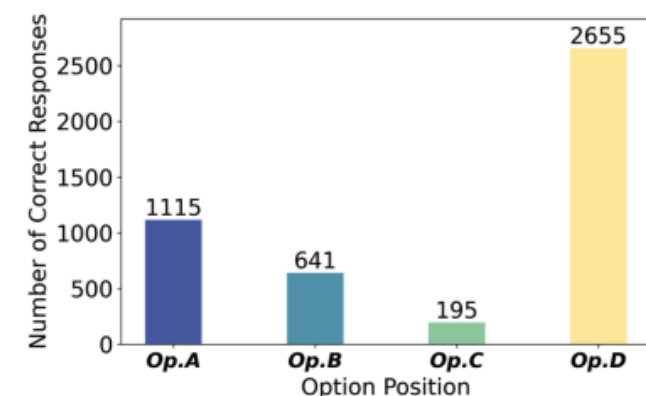
shuffling the order of 4 examples
can lead to 10%~20% fluctuation in
accuracy



(a) Statistics of correct responses for IDEFICS-v2-8B-Instruct in the text-only task of order sensitivity.



(b) Statistics of correct responses for IDEFICS-v2-8B-Instruct in the image-only task of order sensitivity.



(c) Statistics of correct responses for IDEFICS-v2-8B-Instruct in the image-text-pair task of order sensitivity.

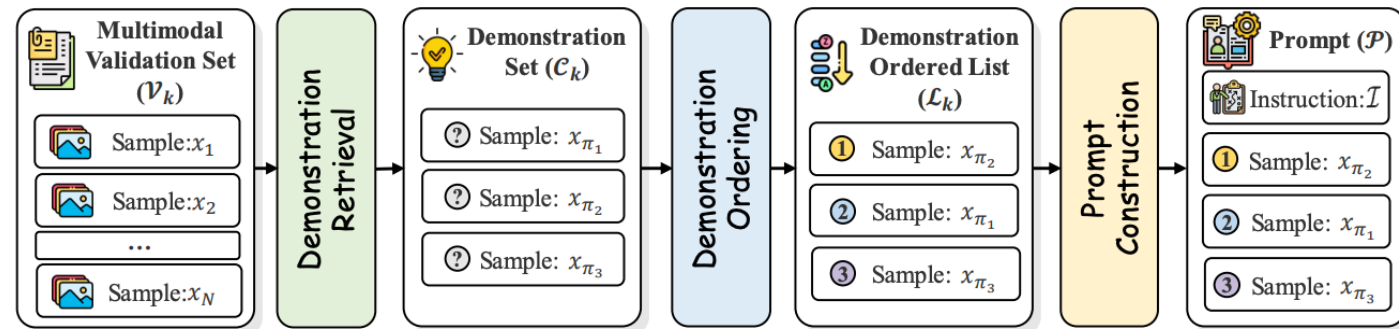
“Lost in the Middle”: Similar to LLM, MLLMs favor first and last position

Open Question: Multimodal In-context Learning

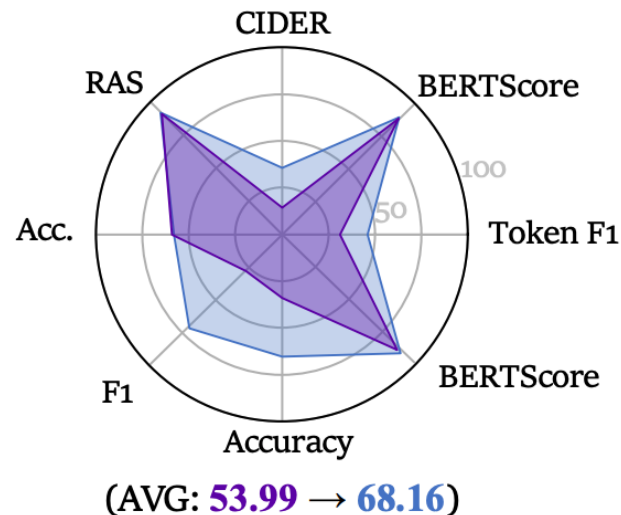
- Why is multimodal ICL difficult?

In-context learning can be **sensitive** to:

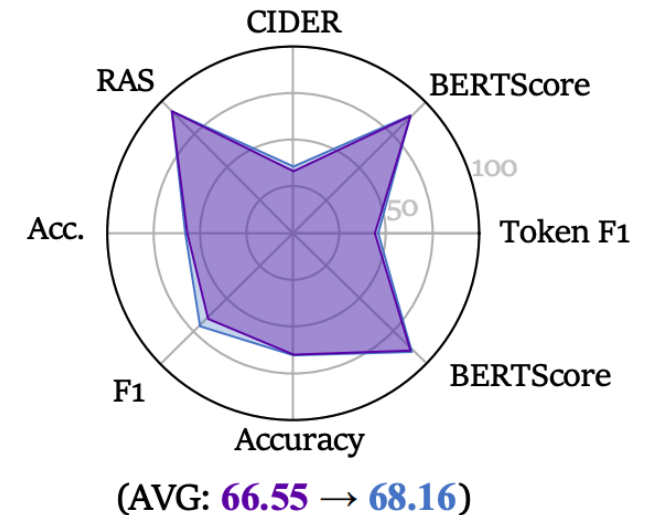
- The order of examples are presented
- The relevance of examples to the target question;
- ...



 L2 Distance
  Cosine Distance

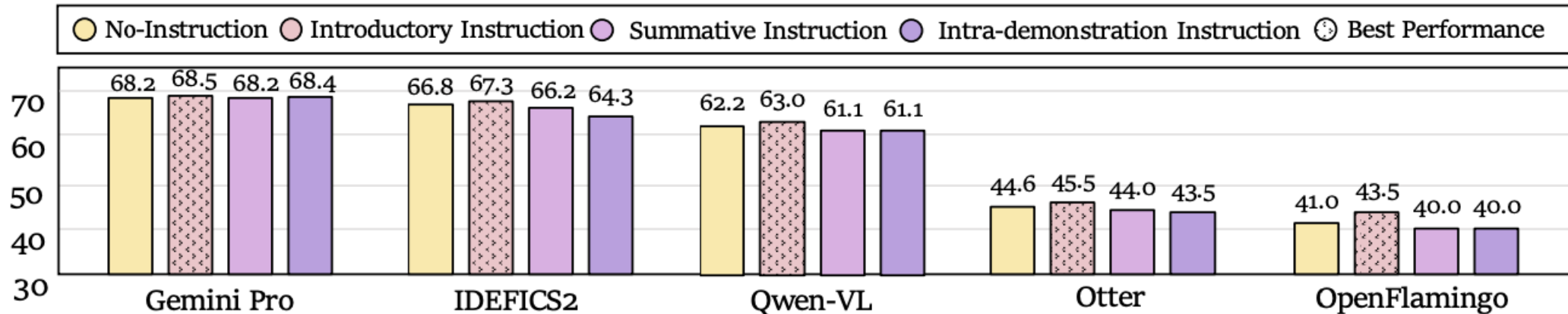
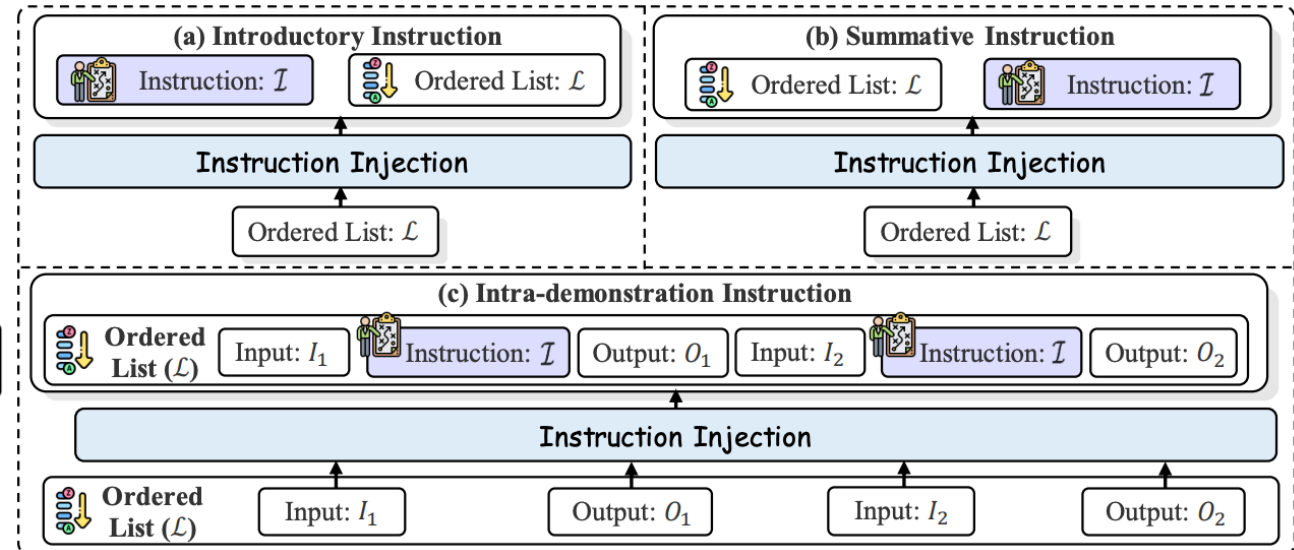
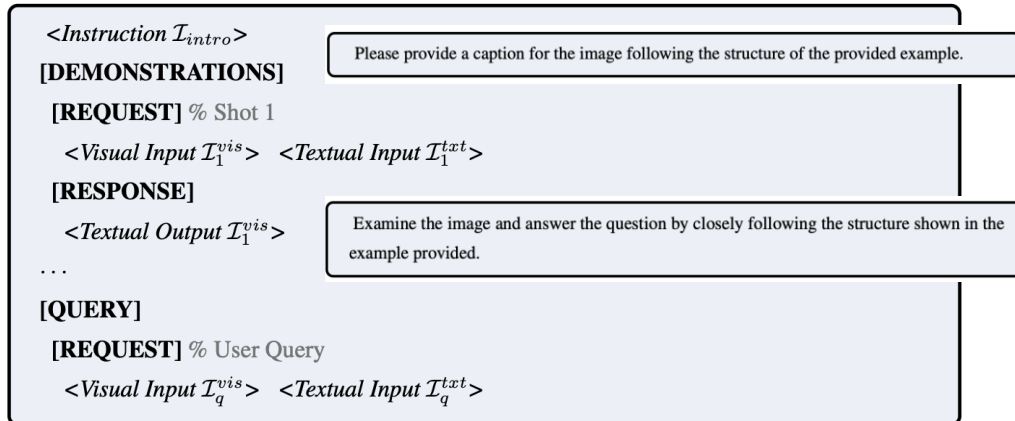


 Diversity Retriever
  Similar Retriever

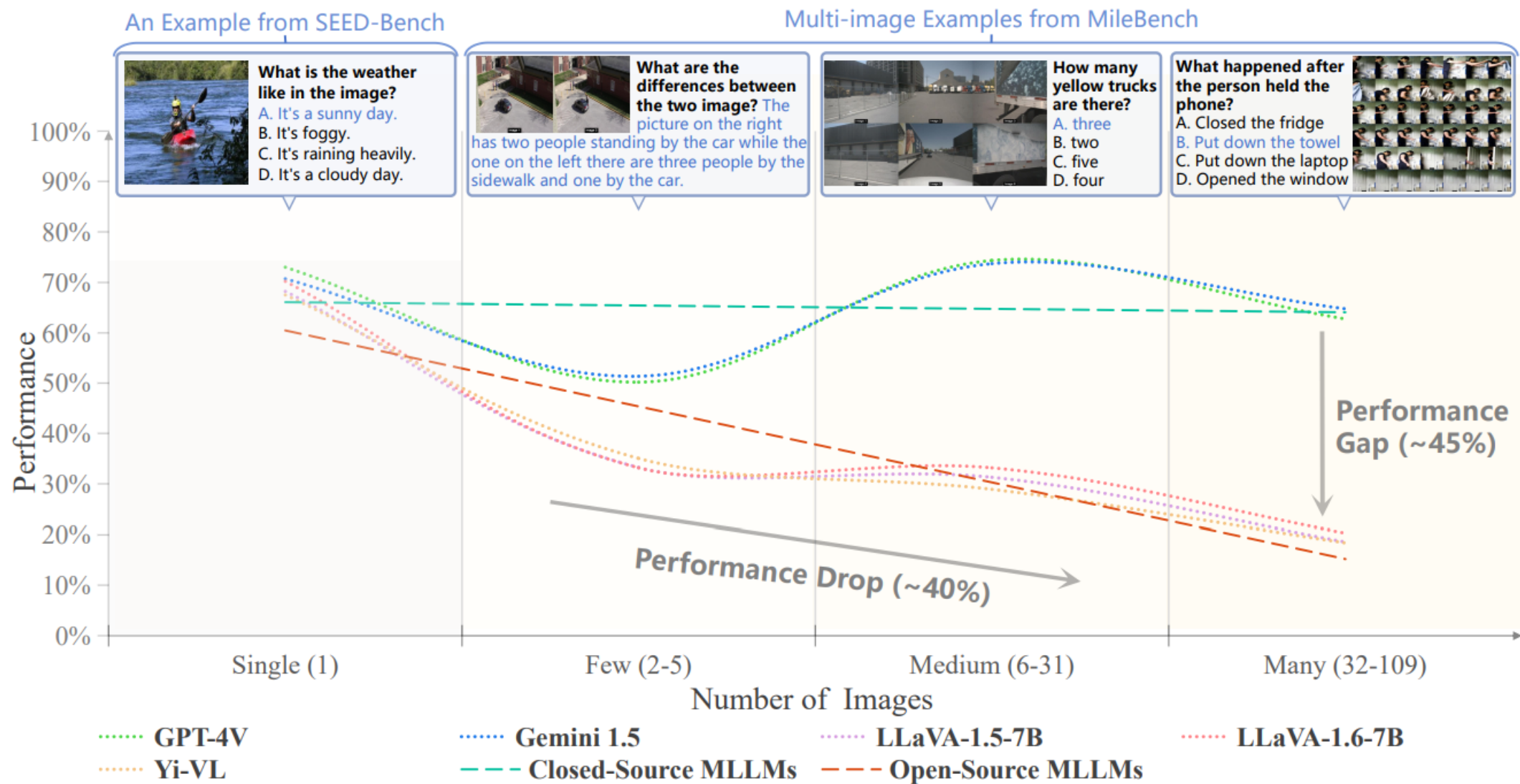


Open Question: Multimodal In-context Learning

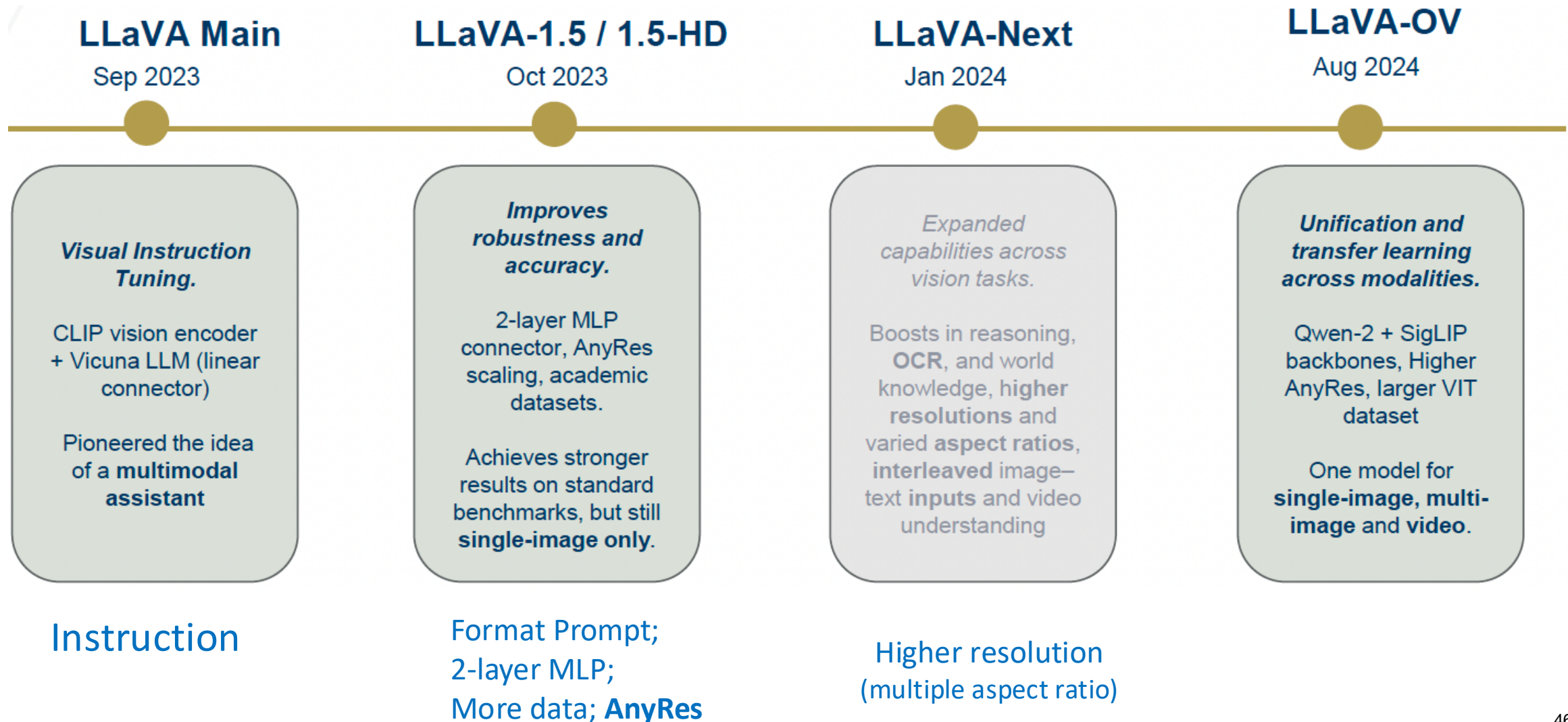
- Introductory instruction matters



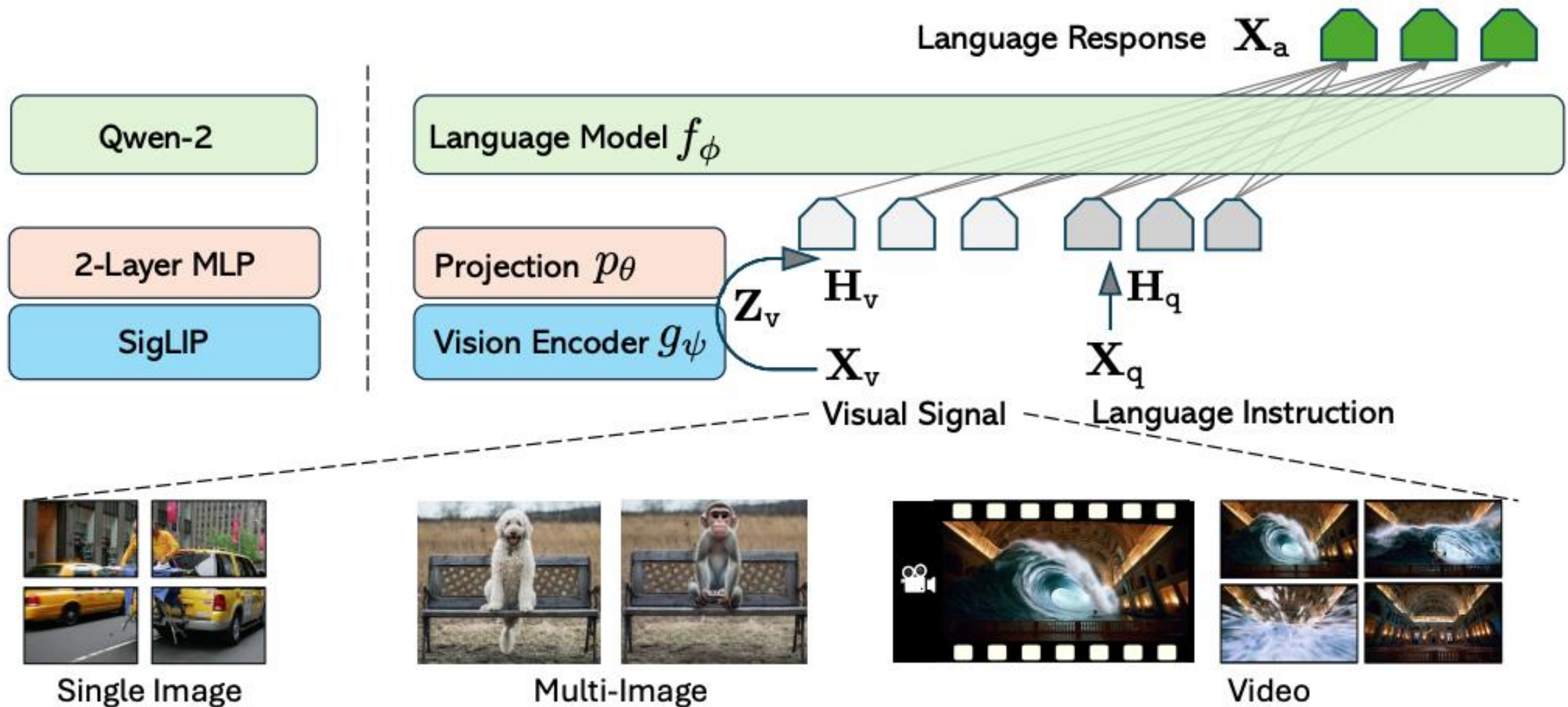
Open Question: Long-context in MLLM



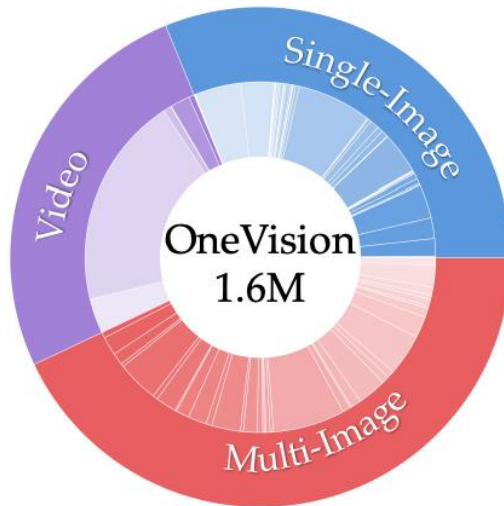
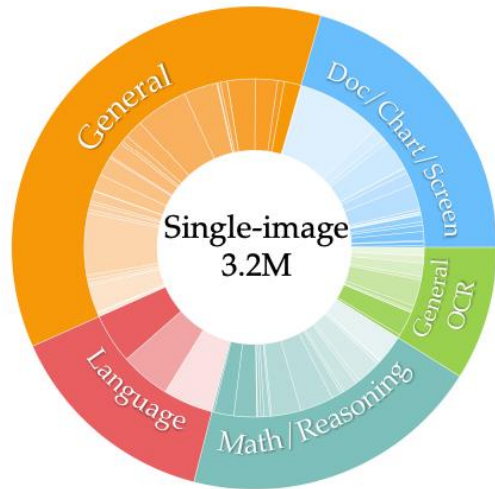
From LLaVA to LLaVA-Onevision



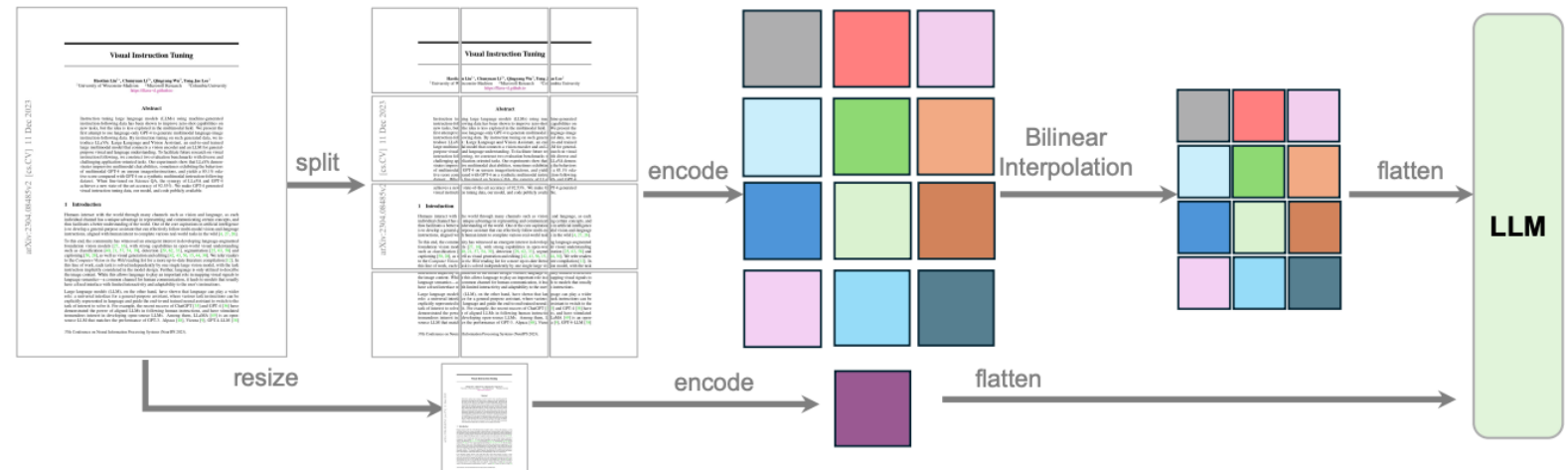
LLaVA-Onevision



LLaVA-Onevision



Single-Image	     ... N Crops	$(1 + 9) * 729 = 7290$ Tokens
Multi-Image	  ... N Images	$12 * 729 = 8748$ Tokens
Video	    ... N Frames	$32 * 196 = 6272$ Tokens
Example on Token Strategy		Max Tokens



(a) Higher AnyRes with Bilinear Interpolation

LLaVA-Onevision

Model	IEI	MI-VQA	NLVR2	Puzzle	Q-Bench	Spot-Diff	TR-VQA	VST	3D-Chat	3D-TD	ScanQA	ALFRED	nuScenes	BLINK	Mantis	MathVerse	MuirBench	SciVerse
	in-domain multi-image								in-domain multi-view					out-domain				
GPT-4V [109]	11.0	52.0	88.8	17.1	76.5	12.5	54.5	10.9	31.2	35.4	32.6	10.3	63.7	51.1	62.7	60.3	62.3	66.9
LLaVA-N-Image-7B [†] [82]	13.2	39.4	68.0	9.0	51.0	12.9	59.6	10.1	-	-	-	-	-	41.8	46.1	13.5	-	12.2
VPG-C-7B [70]	15.2	46.8	73.2	2.4	57.6	27.8	38.9	21.5	-	-	-	-	-	43.1	52.4	24.3	-	23.1
Mantis-7B [47]	11.2	52.5	87.4	25.7	69.9	17.6	45.2	12.5	2.60	14.7	16.1	14.0	46.2	46.4	59.5	27.2	36.1	29.3
LLaVA-N-Inter-7B [68]	24.3	87.5	88.8	48.7	74.2	37.1	76.1	33.1	-	-	-	-	-	52.6	62.7	32.8	38.9	31.6
LLaVA-N-Inter-14B [68]	24.5	95.0	91.1	59.9	76.7	40.5	78.6	33.3	70.6	52.2	34.5	62.0	76.7	52.1	66.4	33.4	40.7	32.7
<i>LLaVA-OV-0.5B (SI)</i>	<i>15.6</i>	<i>44.8</i>	<i>56.1</i>	<i>30.0</i>	<i>45.8</i>	<i>8.5</i>	<i>36.7</i>	<i>7.6</i>	<i>22.1</i>	<i>22.1</i>	<i>16.9</i>	<i>25.5</i>	<i>8.2</i>	<i>37.9</i>	<i>38.2</i>	<i>20.9</i>	<i>22.7</i>	<i>26.7</i>
LLaVA-OV-0.5B	17.1	48.7	63.4	35.4	48.8	36.4	65.0	29.8	60.0	48.0	29.4	62.2	70.5	52.1	39.6	60.0	25.5	29.1
<i>LLaVA-OV-7B (SI)</i>	<i>20.5</i>	<i>60.3</i>	<i>75.9</i>	<i>24.6</i>	<i>56.0</i>	<i>7.9</i>	<i>52.8</i>	<i>8.4</i>	<i>24.5</i>	<i>29.9</i>	<i>22.1</i>	<i>32.0</i>	<i>70.8</i>	<i>45.6</i>	<i>54.2</i>	<i>26.3</i>	<i>32.7</i>	<i>30.0</i>
LLaVA-OV-7B	22.2	90.2	89.4	53.3	74.5	39.2	80.1	31.7	62.8	52.6	30.1	61.0	79.8	48.2	64.2	67.6	41.8	79.1
<i>LLaVA-OV-72B (SI)</i>	<i>22.1</i>	<i>61.2</i>	<i>78.9</i>	<i>44.2</i>	<i>61.5</i>	<i>15.6</i>	<i>67.9</i>	<i>12.1</i>	<i>30.8</i>	<i>25.4</i>	<i>21.9</i>	<i>43.5</i>	<i>75.5</i>	<i>46.0</i>	<i>56.8</i>	<i>58.6</i>	<i>33.2</i>	<i>65.8</i>
LLaVA-OV-72B	22.5	95.3	93.8	63.4	83.2	43.3	83.7	34.5	63.2	53.3	35.8	66.3	78.8	55.4	77.6	91.6	54.8	94.9

LLaVA-Onevision

- S1: Joint understanding of diagram and chart** (single-image to multi-image)
- S2: GUI for multi-modal agent** (single-image and multi-image).
- S3: Set-of-mark Prompting** (Transfer from single-image task composition).
- S4: Image-to-Video Editing Instruction** (from single-image and video).
- S5: Video-to-Video Difference** (Transfer from multi-image and video)
- S6: Multi-camera Video Understanding in Self-driving** (single-image and multi-image to video).
- S7: Composed Sub-video Understanding** (Transfer from multi-image to video).
- S8: Visual prompting in video** (Task transfer from single-image to video).
- S9: Visual Referring in Image in Video Understanding.**

LLaVA-Onevision

Qwen3-vl-8B

S3: Set-of-mark Prompting



User

Here's an image with objects labeled in numbers. Please describe the objects that are marked 5 and 7.

LLaVA-OV

The image shows a room with various items marked with numbers. Here's the description of items marked as 4, 5, and 7:

The item marked as 4 is a framed picture hanging on the wall. It features a black and white illustration of a dandelion with seeds blown away, creating a sense of movement. The item marked as 5 is a white bookshelf with multiple shelves. It appears to be a storage unit that is part of the room's furniture. It is located between the two desks. The item marked as 7 is a black chair with a modern design. It has a metal frame and a woven seat, suggesting a contemporary style. This chair is positioned in front of one of the desks.

IMAGE



INSTRUCTION

Describe Objects Interesting Read text
Depth ELI5

which piece of nugget looks more delicious?

Send

RESPONSE

The image shows nine chicken nuggets arranged to resemble a **"world map"** (each numbered 1–9 to represent continents/regions). While subjective, the most commonly cited "most delicious" piece in this meme is **"#8"** (the nugget shaped like the Middle East).

Why #8?

- It's a **"solid, elongated piece"** with a crisp, golden-brown texture (not broken or overly lumpy like some others).
- Its shape and even browning make it visually appealing in the meme's context, where humor often plays up "the best piece" through regional stereotypes (e.g., the Middle East-shaped nugget being the "most desirable").
- While other pieces (like #3 or #6) might look tasty, #8 is the **"classic meme answer"** for "which nugget looks most delicious."

So, if you're referencing the meme's cultural context, **"#8"** is the go-to choice! 🍗

LLaVA-Onevision

S5: Video-to-Video Difference

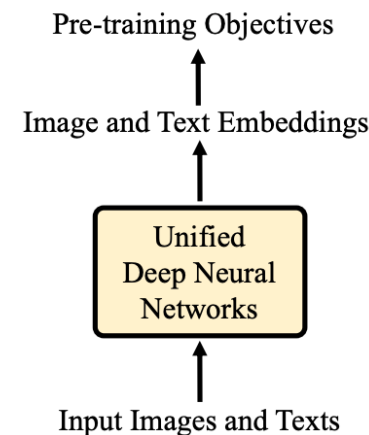


User Here are two videos, please provide the analysis on their differences.

LLaVA-OV The first video shows a sequence of images where a man in a blue plaid shirt approaches a woman in a red dress, engages in a conversation, and eventually gives her a kiss on the cheek. The woman in the red dress is looking at the man with a slight smile, and another woman in a light blue top is observing the scene. The setting appears to be a street with buildings in the background. The second video shows the same man in the blue plaid shirt walking away from the woman in the red dress, who is now looking in the opposite direction. The woman in the light blue top is no longer visible in this sequence. The man is walking down the street, passing by the buildings and other pedestrians. The overall mood of the video changes from the intimate and personal interaction in the first video to a more solitary and neutral scene in the second video.

Up to this point, models like Qwen-VL and LLaVA-OneVision have conquered the spatial and temporal dimensions...

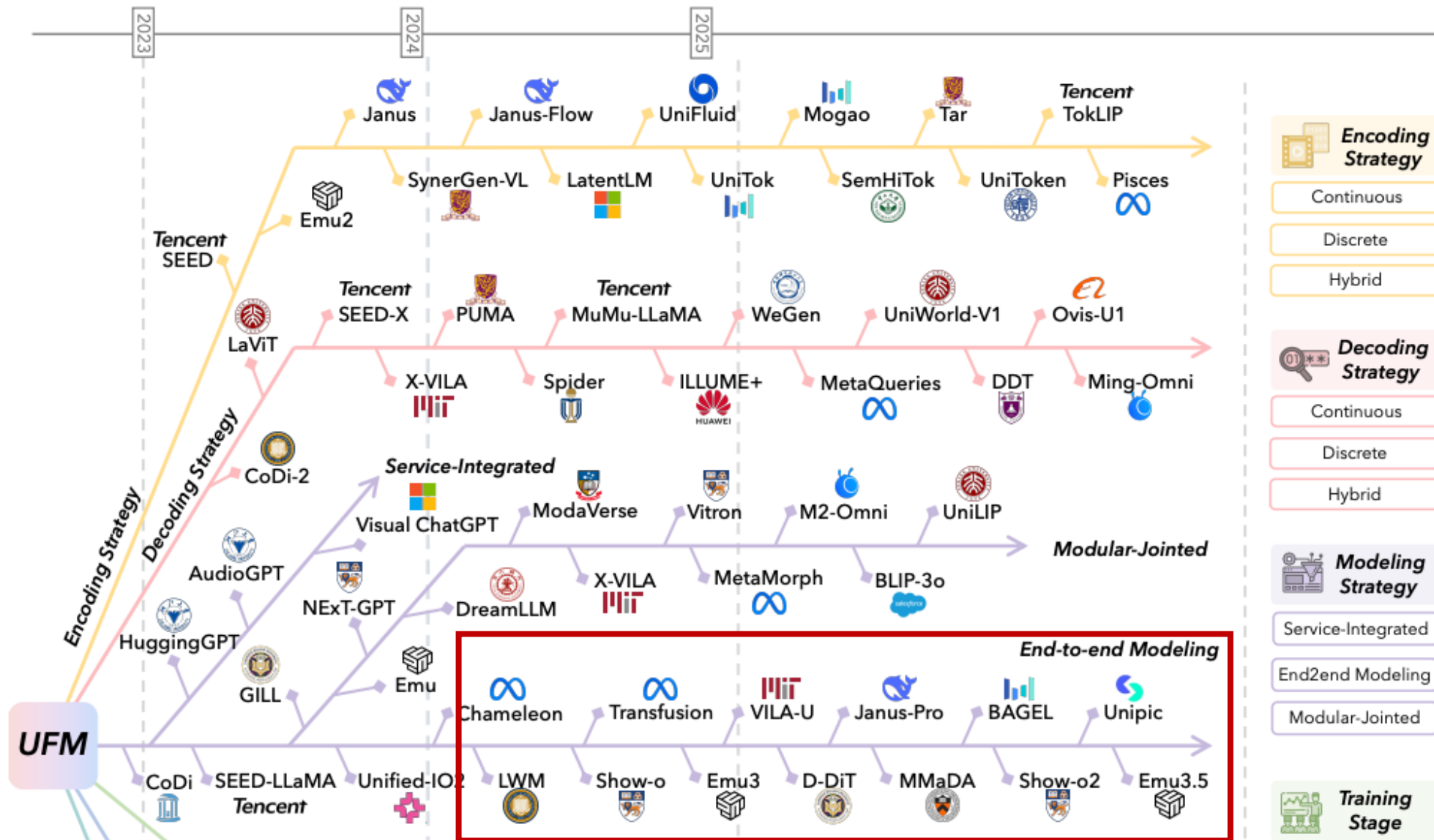
They can only 'read' pixels; they cannot 'write' them.



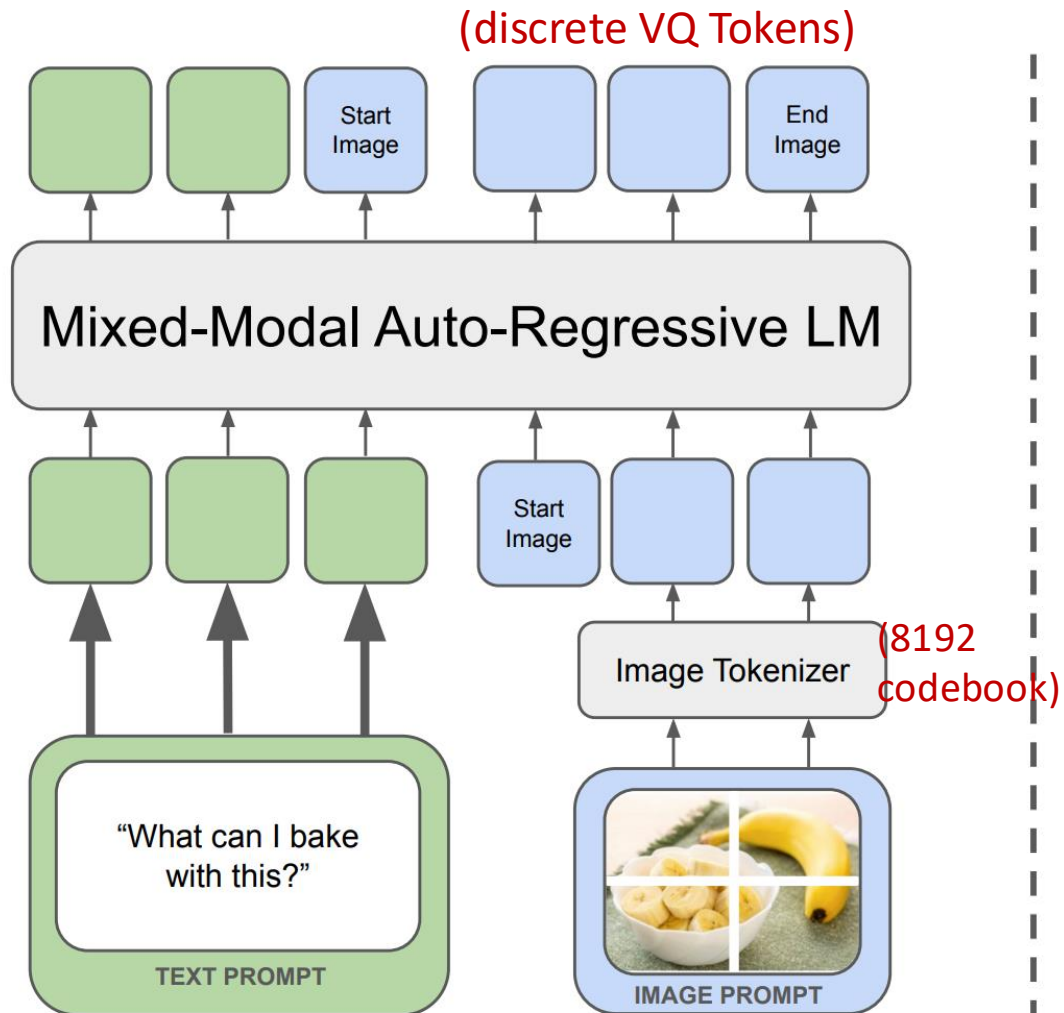
(c) One-Tower VLM

Truly **omni-modal foundation model**

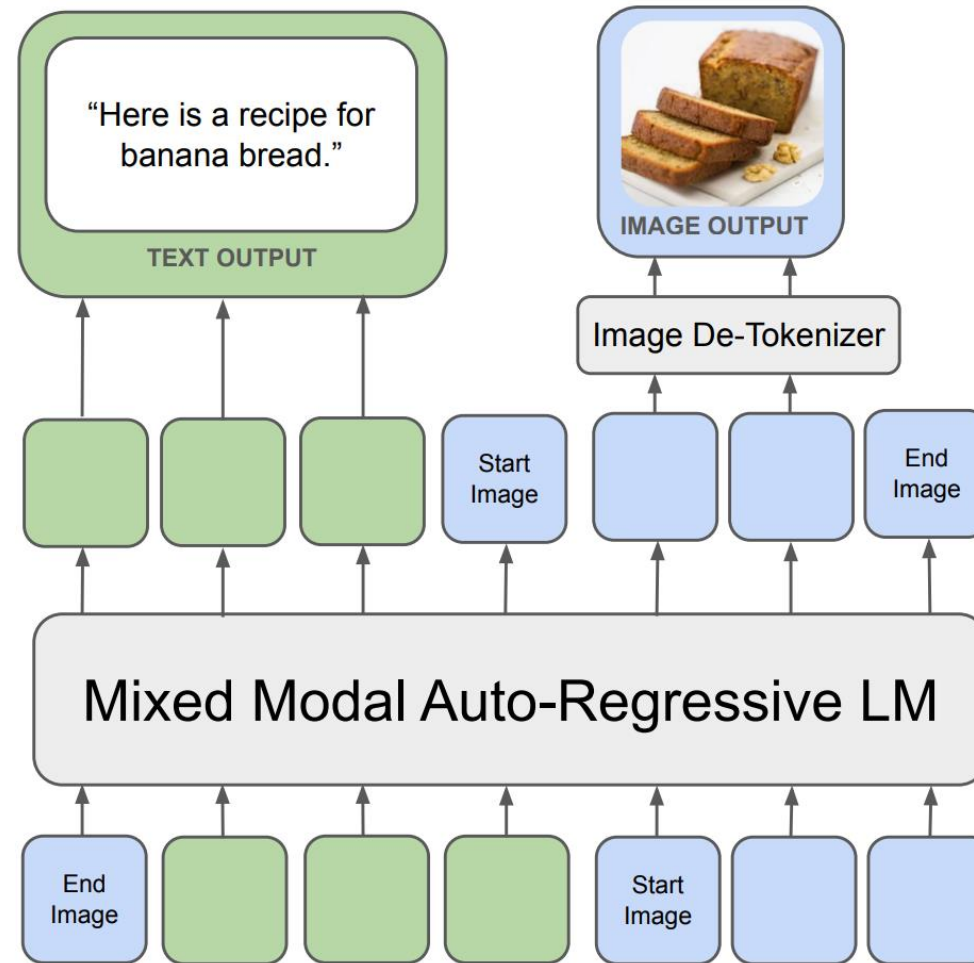
Unified Foundation Models



Chameleon



(a) Mixed-Modal Pre-Training



(b) Mixed-Modal Generation

Chameleon

Advice: 10.2%

What does a meningitis rash look like? What are the other symptoms I should be on the lookout for?

How-to: 12.5%

How do I properly clean my TV screen? I used Windex and now there are towel fibers and wipe marks all over. Show me some reference photos.

Explanation: 14.4%

I've been studying classical French art, and my favorite so far is his painting seen here: **** Could you please give me a few images of other contemporary artworks that have this same aesthetic?



Hypothetical: 5.6%

What would the modern-day vehicle look like if oil had never been discovered?

Brainstorming: 18.6%

Show me a Middle Eastern alternative to these dishes. **<img1> <img2>**



Article: 3.1%

Write me an introduction to a story about knick-knacks, and finish the story by shifting the focus with an image.

Story: 3.9%

Can you create and illustrate a short story for children about an octopus that can't stop eating pizza?

Identification: 9.3 %



Is the below image a Shetland Pony? If not, what is it, and can you show me a Shetland Pony? ****

Comparison: 9.6%

Please tell me what the difference between these two creatures is, and show me some more examples. **<img1> <img2>**



Report: 5.4%

Who designed the church in the image below, and what's the name of the Church? **** Can you please provide me with additional photos of famous landmarks designed by the same architect?



Other: 5.2%

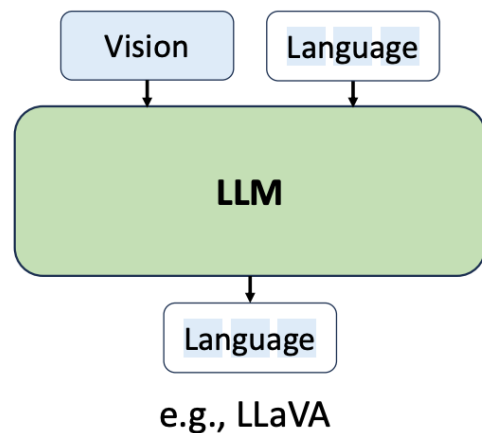
Create a decal for my truck that features running horses as well as the TRD insignia. Use black to gray gradients.

Reasoning: 2.1%

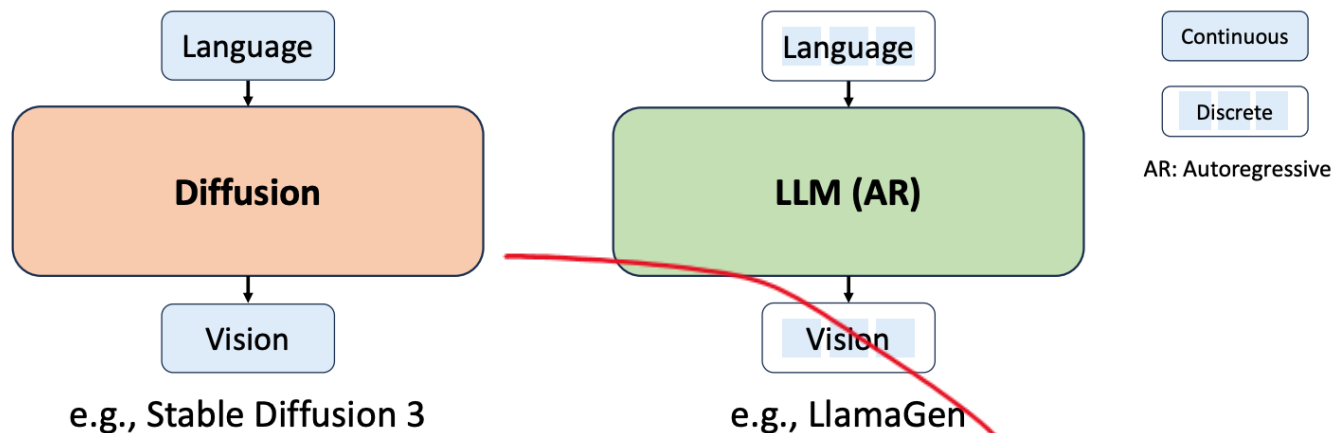
What is typically found at a construction site? Show me a construction site that has a crane.

Show-o

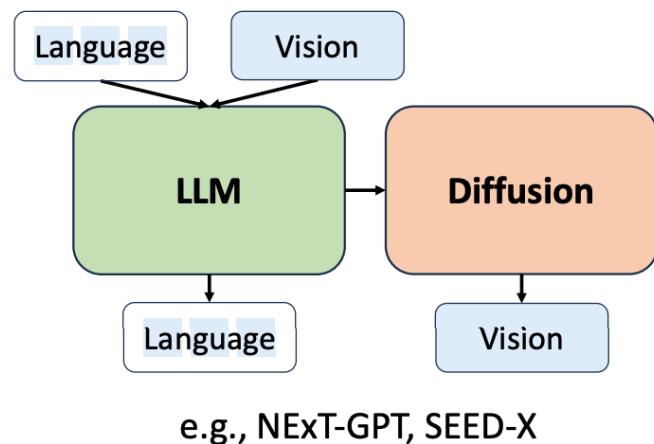
(a) Understanding Only



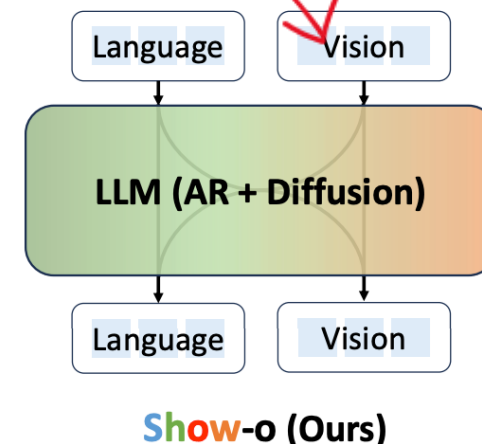
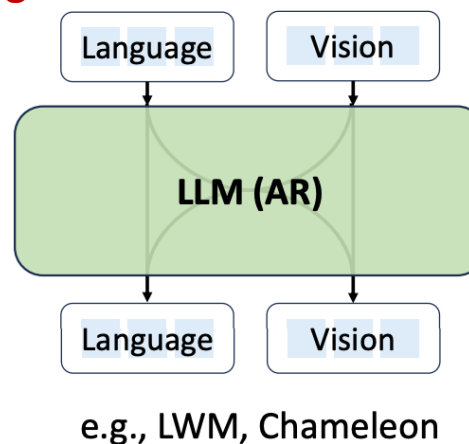
(b) Generation Only



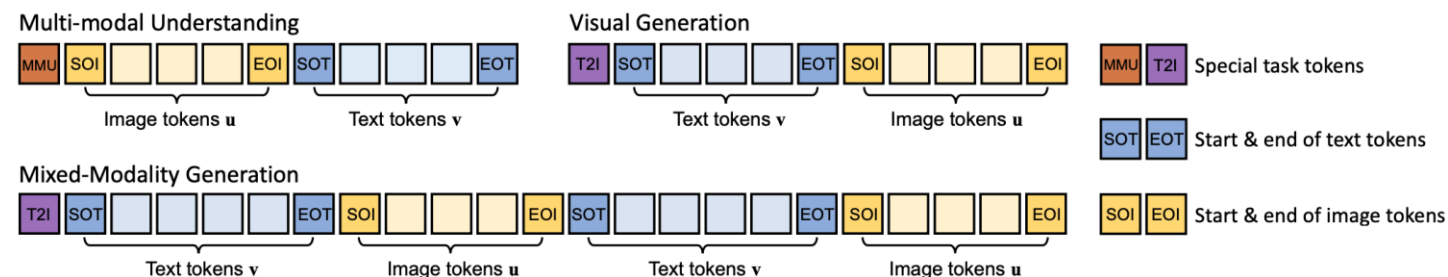
(c) Unified Model (Understanding & Generation)



Next-token prediction for image generation can be less effective...



Show-o



Multimodal Understanding (Captioning, VQA ...)



Q1: Please describe this image in detail.
Q2: Is there a rainbow in this image?

Visual Generation (Text-to-Image Generation / Text-guided Inpainting and Extrapolation)

A punk rock frog in a studded leather jacket shouting into a microphone while standing on a boulder.



a dog sitting on the bench.



a vibrant hot air balloon floats over a clear lake.

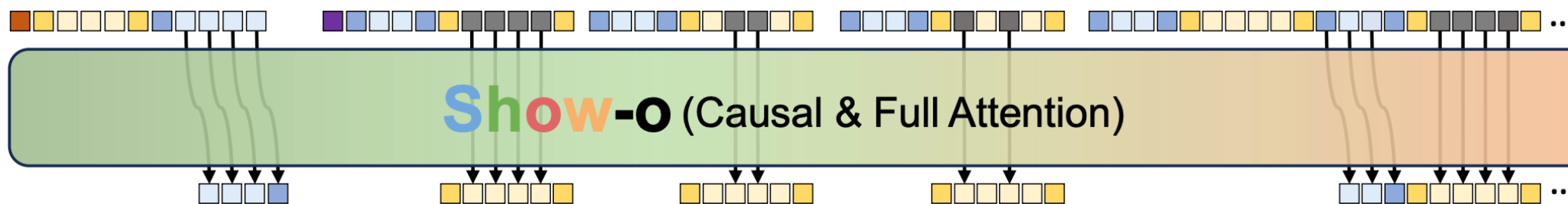


Mixed-modality generation (Video keyframe generation with text descriptions)

Slicing avocado.



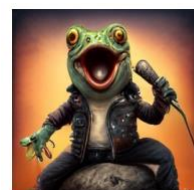
Text Tokenizer & Image Tokenizer



Text De-Tokenizer & Image De-Tokenizer

A1: The image features a young girl sitting on the grass, surrounded by a colourful backdrop. She is holding a ...

A2: Yes, there is a rainbow in the image, as the girl is painting a rainbow on the canvas.



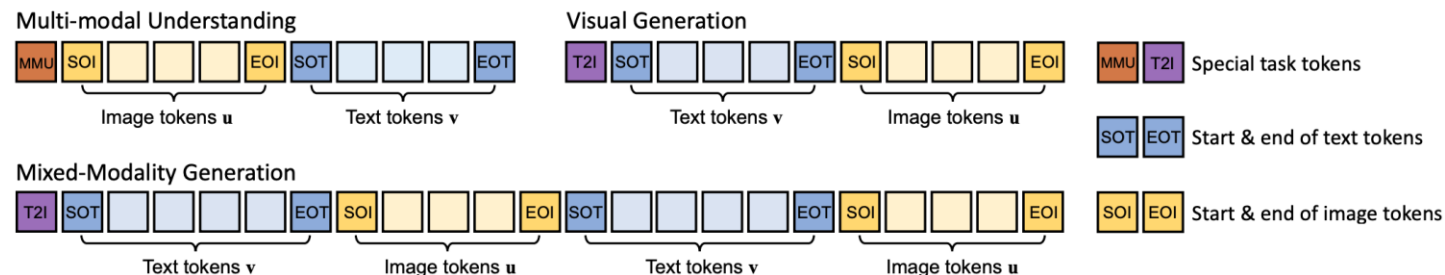
a woman is cutting an avocado with a knife ...



sliced avocado on a white plate.



Janus-Pro



Multimodal Understanding (Captioning, VQA ...)



Q1: Please describe this image in detail.
Q2: Is there a rainbow in this image?

Visual Generation

(Text-to-Image Generation / Text-guided Inpainting and Extrapolation)

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(Video keyframe generation with text descriptions)

Slicing avocado.



Text Tokenizer & Image Tokenizer

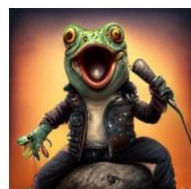
(discrete tokenizer can loss lots of details!)

Show- $\bigcirc$ (Causal & Full Attention)

Text De-Tokenizer & Image De-Tokenizer

A1: The image features a young girl sitting on the grass, surrounded by a colourful backdrop. She is holding a ...

A2: Yes, there is a rainbow in the image, as the girl is painting a rainbow on the canvas.



a woman is cutting an avocado with a knife ...

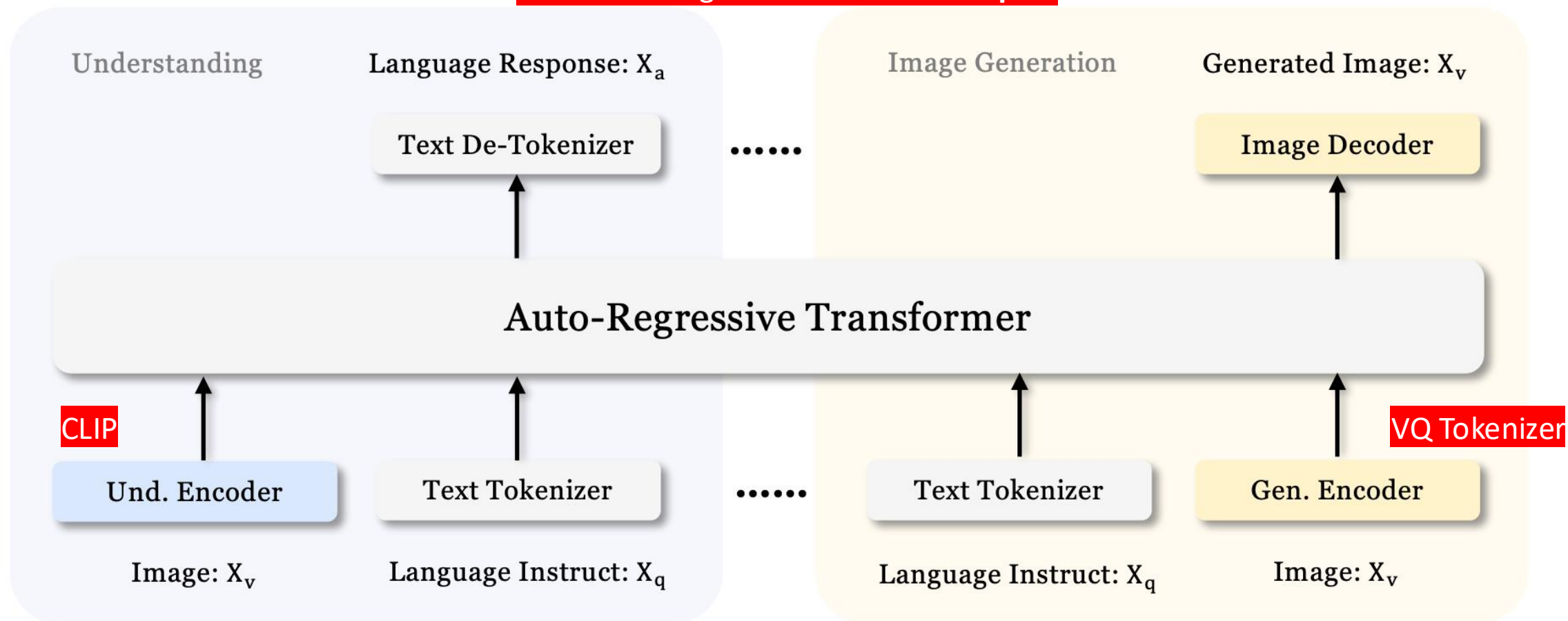


sliced avocado on a white plate.



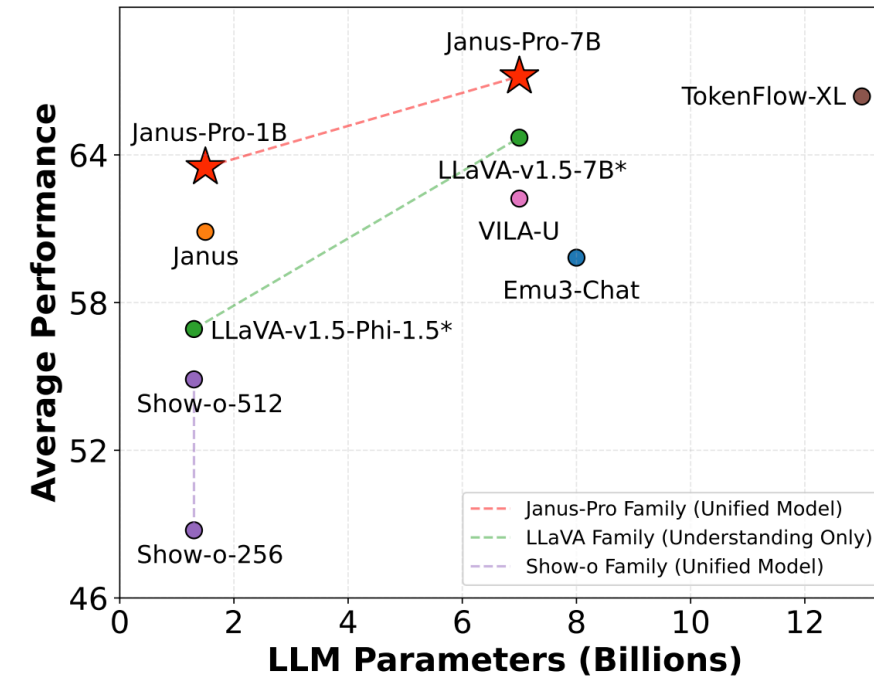
Janus-Pro

Understanding & Generation Decoupled



Janus-Pro

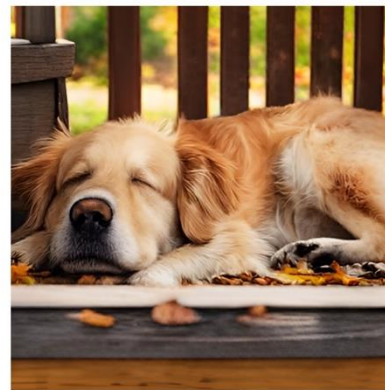
Type	Model	# LLM Params	POPE↑	MME-P↑	MMB↑	SEED↑	GQA↑	MMMU↑	MM-Vet↑
Und. and Gen.	DreamLLM [†] [10]	7B	-	-	-	-	-	-	36.6
	LaVIT [†] [18]	7B	-	-	-	-	46.8	-	-
	MetaMorph [†] [42]	8B	-	-	75.2	71.8	-	-	-
	Emu [†] [39]	13B	-	-	-	-	-	-	-
	NExT-GPT [†] [47]	13B	-	-	-	-	-	-	-
	Show-o-256 [50]	1.3B	73.8	948.4	-	-	48.7	25.1	-
	Show-o-512 [50]	1.3B	80.0	1097.2	-	-	58.0	26.7	-
	D-Dit [24]	2.0B	84.0	1124.7	-	-	59.2	-	-
	Gemini-Nano-1 [41]	1.8B	-	-	-	-	-	26.3	-
	ILLUME [44]	7B	88.5	1445.3	65.1	72.9	-	38.2	37.0
	TokenFlow-XL [34]	13B	86.8	1545.9	68.9	68.7	62.7	38.7	40.7
	LWM [28]	7B	75.2	-	-	-	44.8	-	9.6
	VILA-U [48]	7B	85.8	1401.8	-	59.0	60.8	-	33.5
	Chameleon [40]	7B	-	-	-	-	-	22.4	8.3
	Janus	1.5B	87.0	1338.0	69.4	63.7	59.1	30.5	34.3
	Janus-Pro-1B	1.5B	86.2	1444.0	75.5	68.3	59.3	36.3	39.8
	Janus-Pro-7B	7B	87.4	1567.1	79.2	72.1	62.0	41.0	50.0



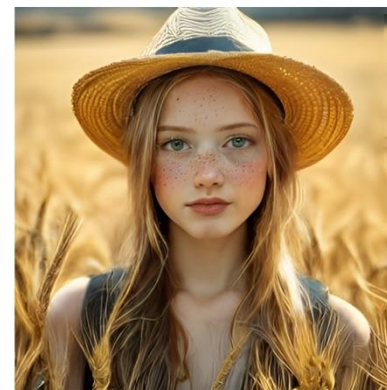
Almost **doubled** the multimodal understanding performance of unified MLLM..

Janus-Pro

Type	Model	# LLM Params	POPE↑	MME-P↑	MMB↑	SEED↑	GQA↑	MMMU↑	MM-Vet↑
<i>Und. and Gen.</i>	DreamLLM [†] [10]	7B	-	-	-	-	-	-	36.6
	LaVIT [†] [18]	7B	-	-	-	-	46.8	-	-
	MetaMorph [†] [42]	8B	-	-	75.2	71.8	-	-	-
	Emu [†] [39]	13B	-	-	-	-	-	-	-
	NExT-GPT [†] [47]	13B	-	-	-	-	-	-	-
	Show-o-256 [50]	1.3B	73.8	948.4	-	-	48.7	25.1	-
	Show-o-512 [50]	1.3B	80.0	1097.2	-	-	58.0	26.7	-
	D-Dit [24]	2.0B	84.0	1124.7	-	-	59.2	-	-
	Gemini-Nano-1 [41]	1.8B	-	-	-	-	-	26.3	-
	ILLUME [44]	7B	88.5	1445.3	65.1	72.9	-	38.2	37.0
	TokenFlow-XL [34]	13B	86.8	1545.9	68.9	68.7	62.7	38.7	40.7
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	Janus-Pro-7B	7B	87.4	1567.1	79.2	72.1	62.0	41.0	50.0



A golden retriever lying peacefully on a wooden porch, with autumn leaves scattered around.



A young woman with freckles wearing a straw hat, standing in a golden wheat field.



A glowing crystal ball floating above a sandstone table in the middle of a desert at sunset.



A tiny galaxy contained inside a glass bottle, glowing brightly against a dark velvet cloth.

Almost **doubled** the multimodal understanding performance of unified MLLM..

Janus-Pro

Try it yourself 😊

<https://huggingface.co/spaces/deepseek-ai/JanusFlow-1.3B>

deepseek-ai / JanusFlow-1.3B like 218 Running on ZERO Get PRO

Hugging Face is way more fun with friends and colleagues! 🥳 [Join an organization](#) Dismiss this message

Multimodal Understanding



Decoupling Visual Encoding



Single Visual Encoder

Question

explain this meme

Seed

42

top_p

0.95

temperature

0.1

Chat

Response

This meme compares the strength of two different visual encoding methods: Decoupling Visual Encoding and Single Visual Encoder. The left side of the meme shows a muscular dog representing Decoupling Visual Encoding, which is a more complex and powerful method that can handle more complex visual data. The right side shows a slouched dog representing Single Visual Encoder, which is a simpler and less powerful method that can only handle a single visual input. The meme humorously suggests that Decoupling Visual Encoding is much stronger than Single Visual Encoder.

Multimodal Understanding examples

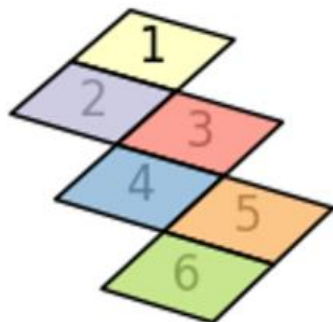
Question	
explain this meme	
Convert the formula into latex code.	

Text-to-Image Generation

CFG Weight Number of Inference Steps

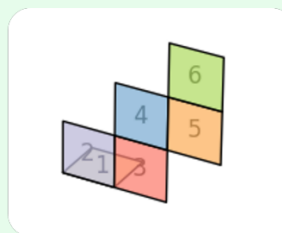
Why Unified MLLM is important?

Question

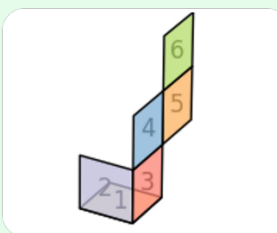


Can the net be folded to form a cube, yes or no?

Step-by-Step Human Mental Simulation



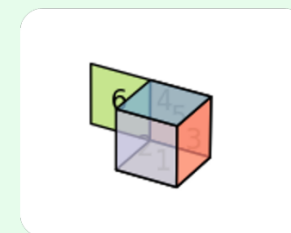
1



2



3



4

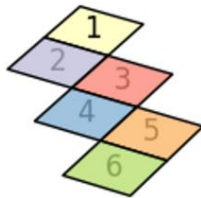


Yes!



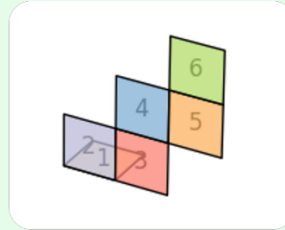
Why Unified MLLM is important?

Question

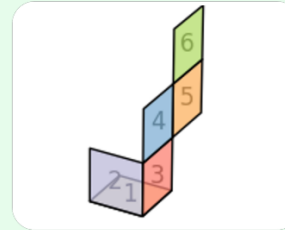


Can the net be folded to form a cube, yes or no?

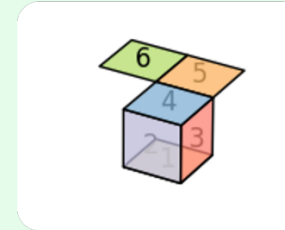
Step-by-Step Human Mental Simulation



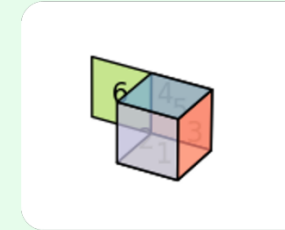
1



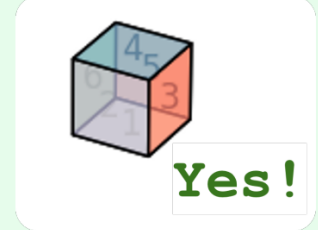
2



3



4



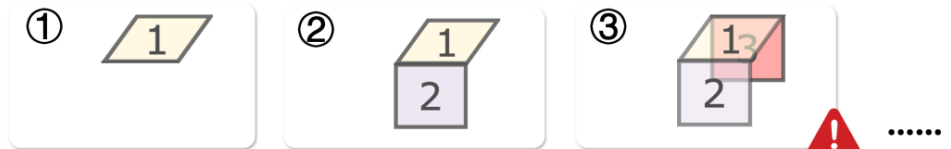
MLLM Textual Simulation

GPT-4o: Yes, this net can be folded into a cube. Here's how:

- ① Face 1 becomes the top of the cube.
- ② Face 2 folds down to the left of Face 1.
- ③ Face 3 folds down to the right of Face 1. } 
- ④ Face 4 folds down below Face 2, becoming the front face.
- ⑤ Face 5 folds down below Face 3, becoming the back face.
- ⑥ Face 6 folds up to close the bottom of the cube.

All faces will perfectly align , forming a six-faced cube with no overlaps or gaps.

Visualization of Textual Reasoning:

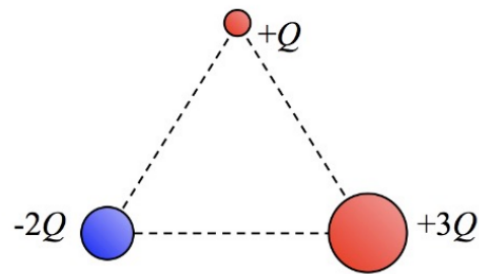


Reasoning Error Analysis: If Face 2 is on the **left** of Face 1 (②) and Face 3 is on the **right** (③), they would be opposite, but they are adjacent in the net. The correct third step should follow the net's layout to maintain their connection.

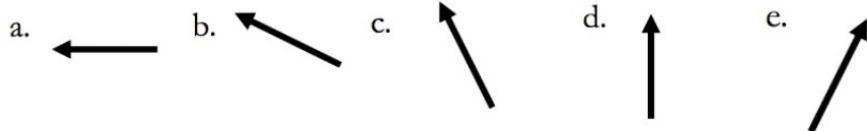
Why Unified MLLM is important?

Visual Simulation is Critical to Human (non-verbal) Reasoning

Multimodal reasoning question in EMMA

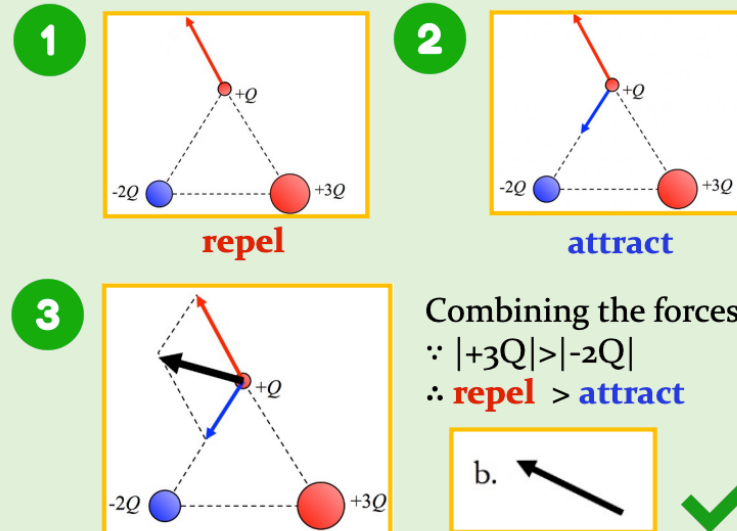


Question: Three point charges, of charge $+Q$, $-2Q$, and $+3Q$, are placed equidistant as shown. Which vector best describes the net direction of the electric force acting on the $+Q$ charge?



Human reasoning

The direction of the electric force due to $-2Q$ and $+3Q$ can be visualized with a quick free-body sketch of the electric forces:



Model reasoning (GPT-4o)

1. Forces Due to the Charges:

- The $-2Q$ charge attracts the $+Q$ charge. (Force points downward-left).
- The $+3Q$ charge repels the $+Q$ charge. (Force points downward-right). **✗**

2. Force Magnitudes (Coulomb's Law):

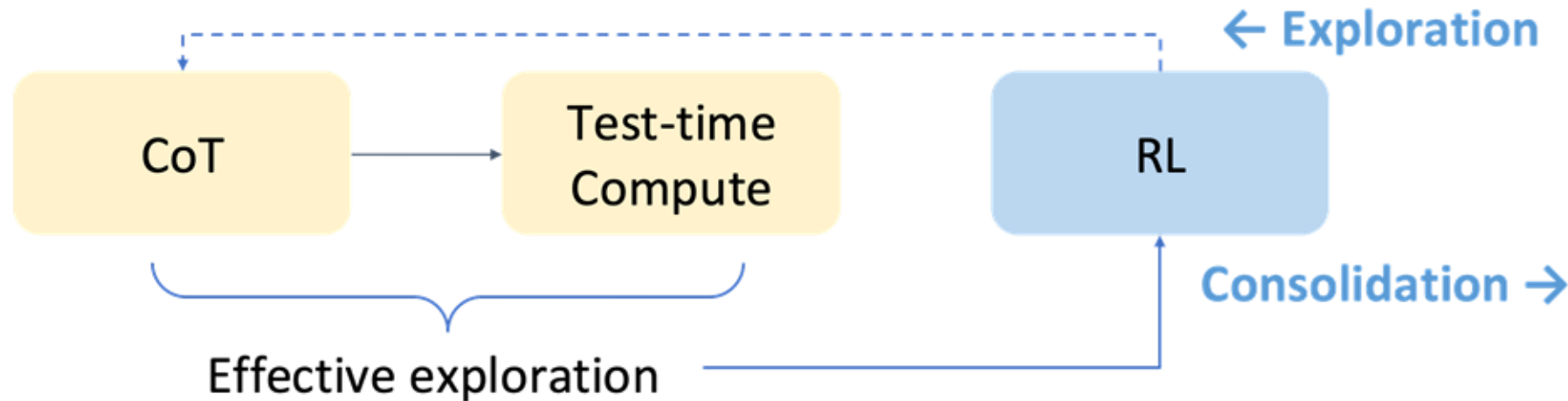
- Force due to $-2Q$: $F_{-2Q} = \frac{2kQ^2}{r^2}$.
- Force due to $+3Q$: $F_{+3Q} = \frac{3kQ^2}{r^2}$.

.....

Error: The issue arose from not correctly identifying the force direction from $+3Q$ to $+Q$, which causes the net force to point ↘, not downward-right↘.

Enabling Models to Think (Visually)

Visual Simulation is Critical to Human (non-verbal) **Reasoning** (will mention more in Week 10)



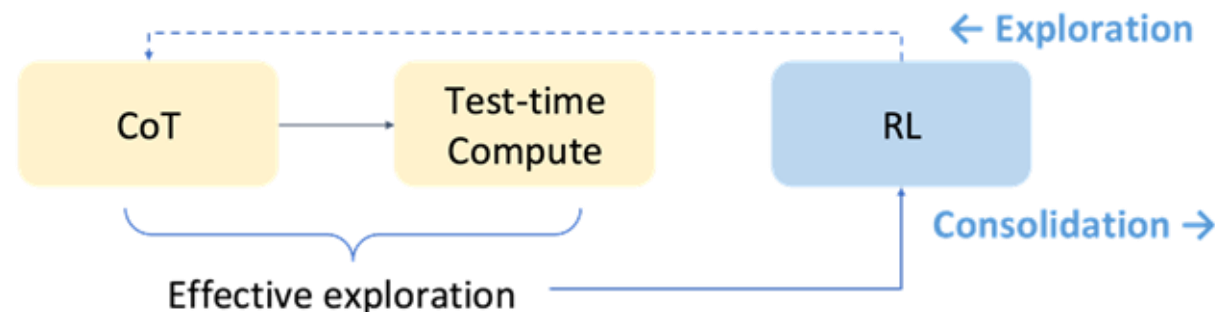
What is Chain-of-Thought (CoT)?

- A method that elicits reasoning in LLM/VLMs by generating a series of intermediate reasoning steps leading to the final answer.

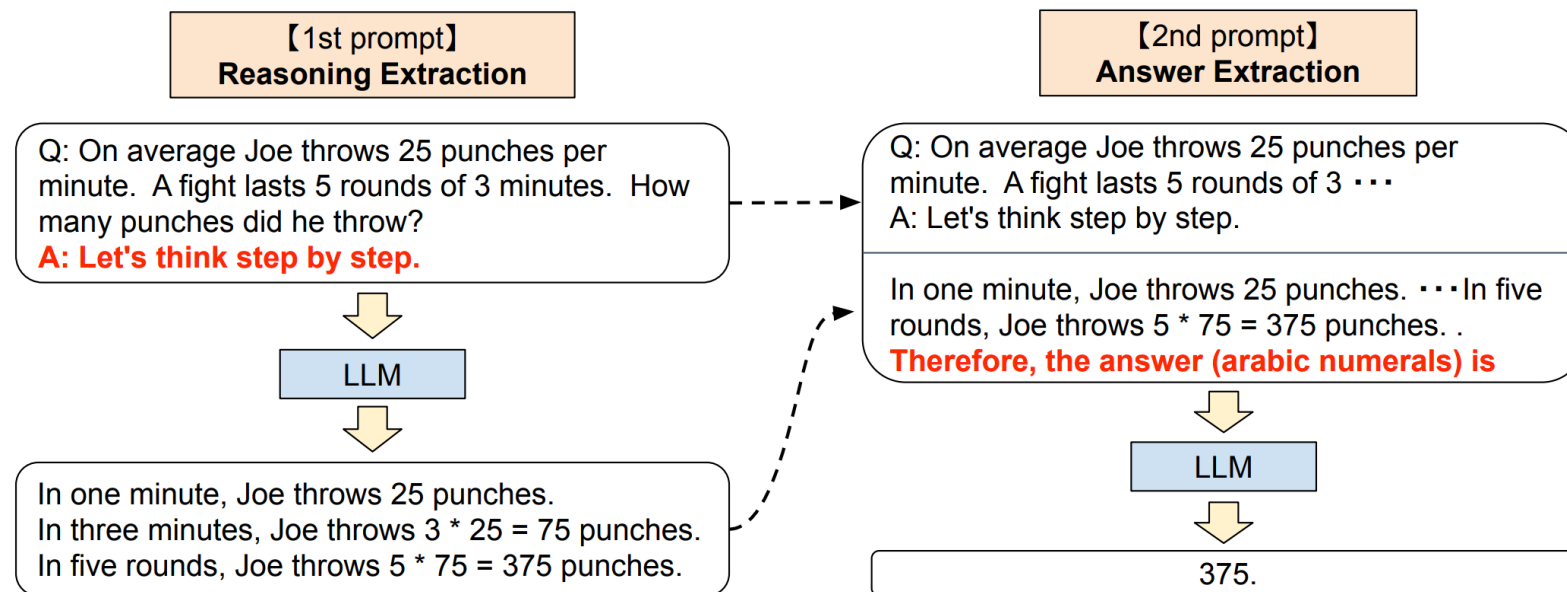
Standard Prompting	Chain-of-Thought Prompting
Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: The answer is 11.	Model Input Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now? A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.
Model Output A: The answer is 27. ❌	Model Output A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Enabling Models to Think (Visually)

Visual Simulation is Critical to Human (non-verbal) **Reasoning** (will mention more in Week 10)

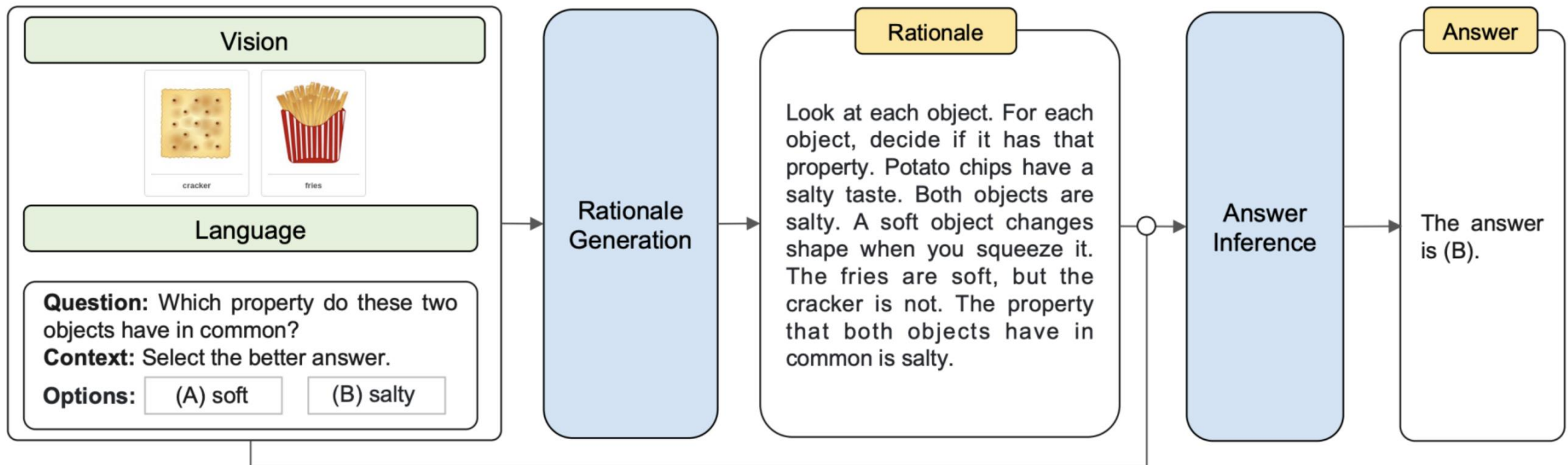


Adding the phrase “Let’s think step by step” before an answer, **Zero-Shot CoT** enables models to perform reasoning step

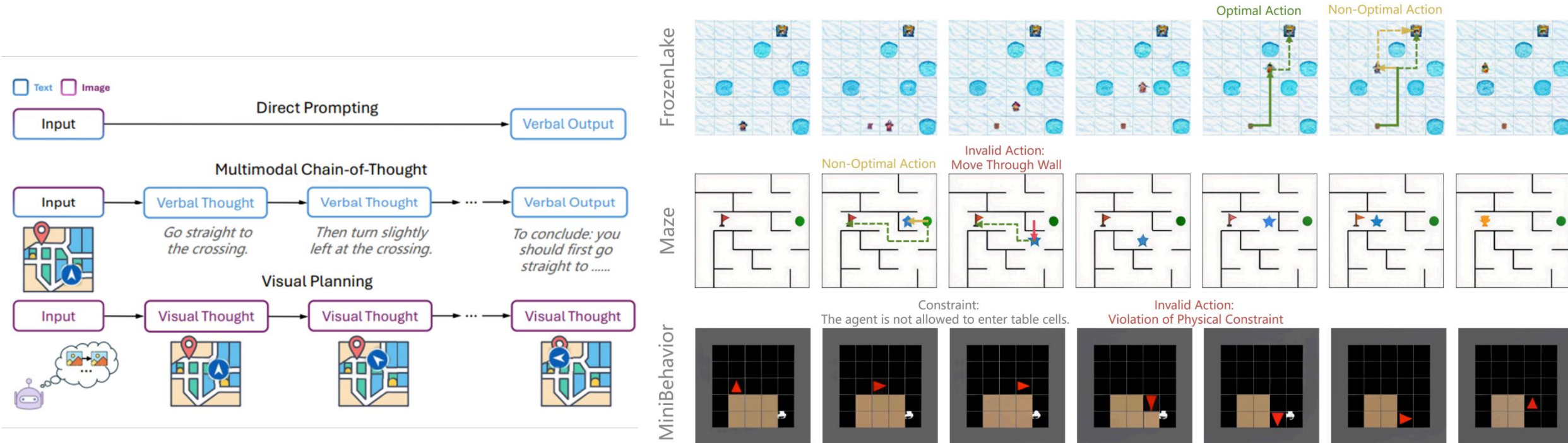


Enabling Models to Think (Visually)

Multimodal Chain-of-Thought Reasoning



Open Question: Enabling Models to Think (**Visually**)



- Some reasoning problems are inherently **spatial** and **state-based**.
- Instead of verbalizing every intermediate step, a model may benefit from **reasoning over visual intermediate states**.
- The open challenge is to make such visual thoughts **structured, faithful, and constraint-aware**, so they improve planning rather than just generate plausible images.

Open Question: World Model

Emu3.5: Native Multimodal Models are World Learners

Emu3.5 Team

BAAI

<https://emu.world>

Abstract

We introduce Emu3.5, a large-scale multimodal world model that natively predicts the next state across vision and language. Emu3.5 is pre-trained end-to-end with a unified next-token prediction objective on a corpus of vision-language interleaved data containing over 10 trillion tokens, primarily derived from sequential frames and transcripts of internet videos. The model naturally accepts interleaved vision-language inputs and generates interleaved vision-language outputs. Emu3.5 is further post-trained with large-scale reinforcement learning to enhance multimodal reasoning and generation. To improve inference efficiency, we propose Discrete Diffusion Adaptation (DiDA), which converts token-by-token decoding into bidirectional parallel prediction, accelerating per-image inference by about 20× without sacrificing performance. Emu3.5 exhibits strong native multimodal capabilities, including long-horizon vision-language generation, any-to-image (X2I) generation, and complex text-rich image generation. It also exhibits generalizable world-modeling abilities, enabling spatiotemporally consistent world exploration and open-world embodied manipulation across diverse scenarios and tasks. For comparison, Emu3.5 achieves performance comparable to Gemini 2.5 Flash Image (Nano Banana) on image generation and editing tasks and demonstrates superior results on a suite of interleaved generation tasks. We open-source Emu3.5 at <https://github.com/baaivision/Emu3.5> to support community research.

Next frames $\mathbf{o}' \sim \hat{T}(\mathbf{o}, \mathbf{a})$ Control actions

Previous frames

- World models infer possible future states of the world from historical observations and alternative actions, and are regarded as one pathway toward autonomous intelligence

Open Question: World Model

Emu3.5: Native Multimodal Models are World Learners

Emu3.5 Team

BAAI

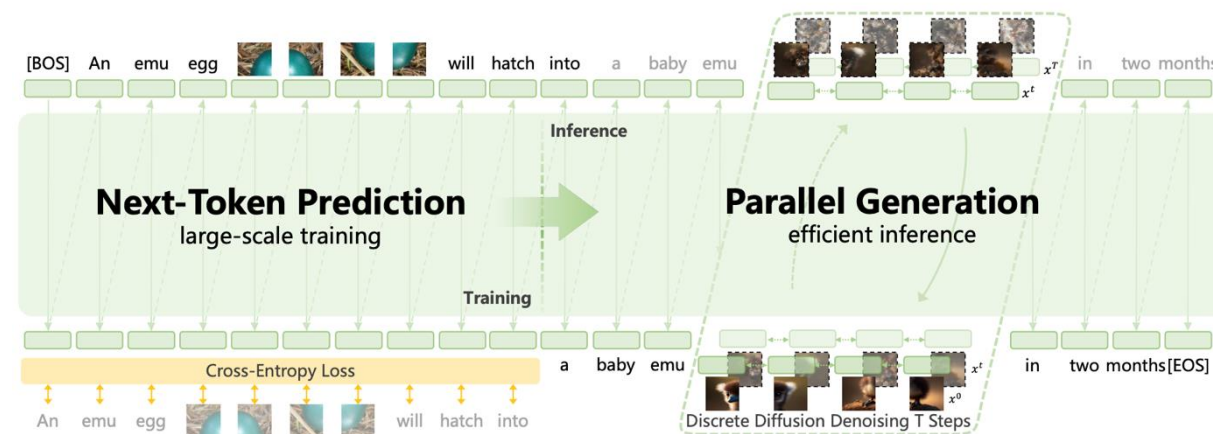
<https://emu.world>

Abstract

We introduce Emu3.5, a large-scale multimodal world model that natively predicts the next state across vision and language. Emu3.5 is pre-trained end-to-end with a unified next-token prediction objective on a corpus of vision-language interleaved data containing over 10 trillion tokens, primarily derived from sequential frames and transcripts of internet videos. The model naturally accepts interleaved vision-language inputs and generates interleaved vision-language outputs. Emu3.5 is further post-trained with large-scale reinforcement learning to enhance multimodal reasoning and generation. To improve inference efficiency, we propose Discrete Diffusion Adaptation (DiDA), which converts token-by-token decoding into bidirectional parallel prediction, accelerating per-image inference by about 20× without sacrificing performance. Emu3.5 exhibits strong native multimodal capabilities, including long-horizon vision-language generation, any-to-image (X2I) generation, and complex text-rich image generation. It also exhibits generalizable world-modeling abilities, enabling spatiotemporally consistent world exploration and open-world embodied manipulation across diverse scenarios and tasks. For comparison, Emu3.5 achieves performance comparable to Gemini 2.5 Flash Image (Nano Banana) on image generation and editing tasks and demonstrates superior results on a suite of interleaved generation tasks. We open-source Emu3.5 at <https://github.com/baaivision/Emu3.5> to support community research.

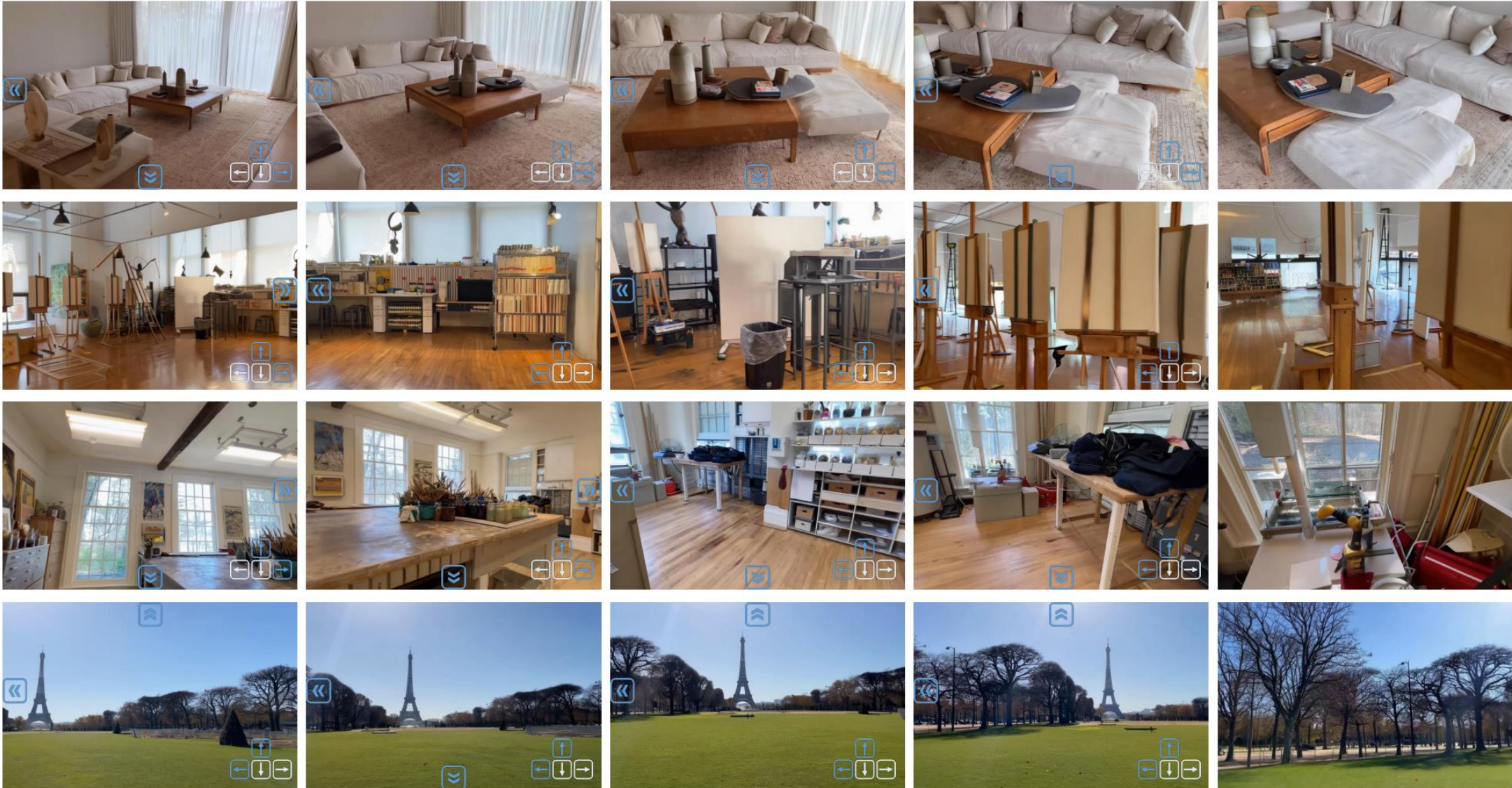
Next frames $\mathbf{o}' \sim \hat{T}(\mathbf{o}, \mathbf{a})$ Control actions

Previous frames

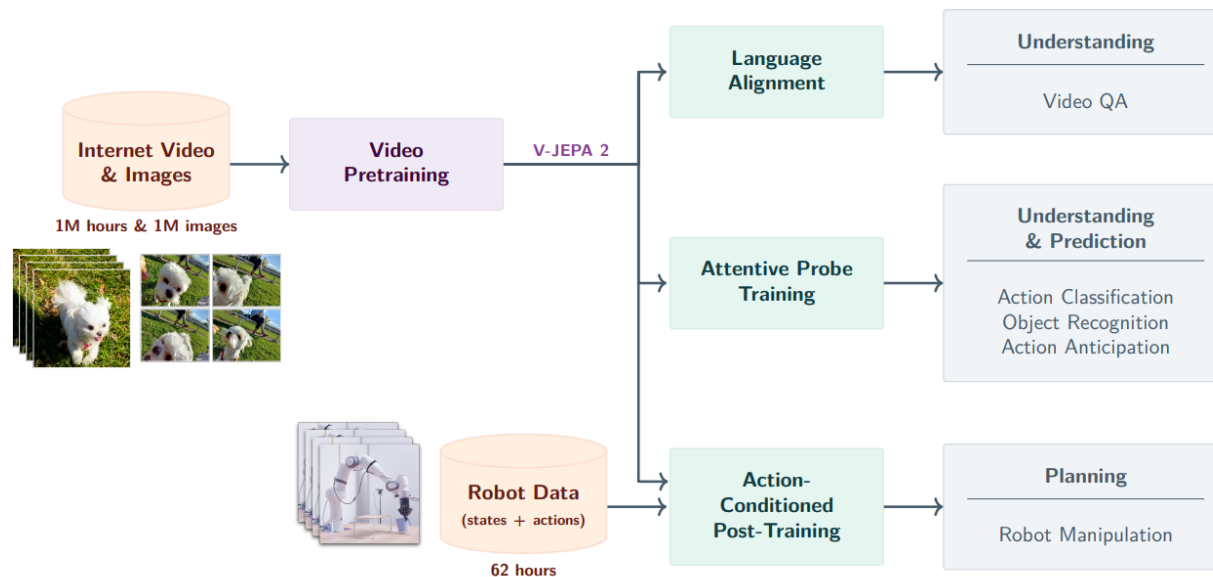


- World models infer possible **future states** of the world from **historical observations** and **alternative actions**, and are regarded as one pathway toward autonomous intelligence

Open Question: World Model



Open Question: World Model



V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning

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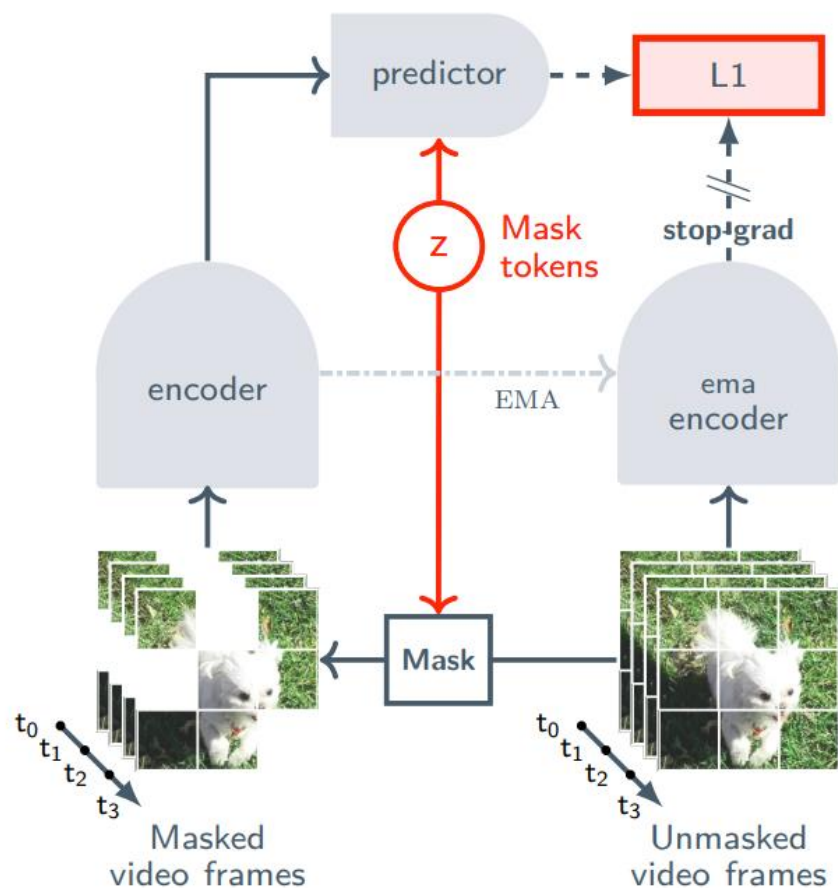
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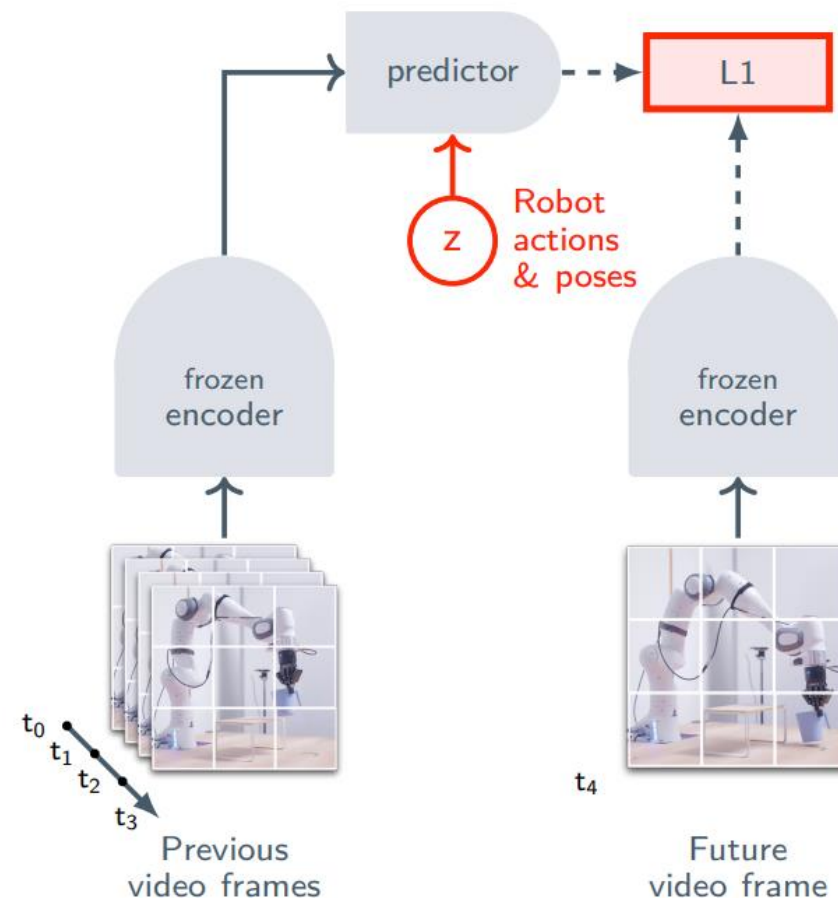
A major challenge for modern AI is to learn to understand the world and learn to act largely by observation (LeCun, 2022). This paper explores a self-supervised approach that combines internet-scale video data with a small amount of interaction data (robot trajectories), to develop models capable of understanding, predicting, and planning in the physical world. We first pre-train an action-free joint-embedding-predictive architecture, V-JEPA 2, on a video and image dataset comprising over 1 million hours of internet video. V-JEPA 2 achieves strong performance on motion understanding (77.3 top-1 accuracy on Something-Something v2) and state-of-the-art performance on human action anticipation (39.7 recall-at-5 on Epic-Kitchens-100) surpassing previous task-specific models. Additionally, after aligning V-JEPA 2 with a large language model, we demonstrate state-of-the-art performance on multiple video question-answering tasks at the 8 billion parameter scale (e.g., 84.0 on PerceptionTest, 76.9 on TempCompass). Finally, we show how self-supervised learning can be applied to robotic planning tasks by post-training a latent action-conditioned world model, V-JEPA 2-AC, using less than 62 hours of unlabeled robot videos from the Droid dataset. We deploy V-JEPA 2-AC zero-shot on Franka arms in two different labs and enable picking and placing of objects using planning with image goals. Notably, this is achieved without collecting any data from the robots in these environments, and without any task-specific training or reward. This work demonstrates how self-supervised learning from web-scale data and a small amount of robot interaction data can yield a world model capable of planning in the physical world.

Open Question: World Model

V-JEPA 2



V-JEPA 2-AC



What we covered today...

After today's class, you should be able to

- trace the evolution from **LLaVA** to **LLaVA-Next**, **Qwen2-VL** and **LLaVA-OneVision**
- understand **multimodal in-context learning** at a high level
- recognize the shift toward **unified multimodal systems**
- open questions: **high-resolution** and **long-context**, **visual thinking**, **world model**

Tutorial (Zhizhen Zhang): Hands-on practice on Multimodal RAG + MLLM

Next Week: Vector Databases (High-Dimensional Indexing)



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THANKS!

Do you have any questions?

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